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**Parametric Aerodynamic Study of the Properties of
Leading-Edge Serrations on Wing Sections**

Thesis submitted as a requirement for the degree of Bachelor of
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Abstract

In this paper, the aerodynamics of various geometric properties of leading-edge serrations on wing sections are explored through automated large-eddy simulation (LES) computational optimisation in ANSYS Fluent, with experimental verification. The wavelength, amplitude, and termination distance of the serrations are parameterised on the cambered NACA2412 airfoil for an aspect-ratio two wing-section. For a Reynolds number of 100,000 and 500,000, the pre-stall, stall, and post-stall regimes are explored, and the relative aerodynamic performance of the serration combinations is found. The study found that in the pre-stall regime, smaller, less perturbing serrations are favoured for their reduced drag, with a slight but unanimous performance reduction compared to the unserrated wing section. As the wing-sections are pushed to stall and post-stall regimes, the opposite is found. The large-amplitude serrations dominate the post-stall regime, which correlates with their ability to create stable vortex structures that persist beyond 50% of the chord. The serration combination that maintains performance across all three regimes at Reynolds number of 100,000 has a wavelength of $1/6.5$, an amplitude of 0.03, and a termination distance of 0.35 (specific to the average chord) with performance at Reynolds number of 500,000 non-uniform across stall regimes. A strong correlation between the wavelength-to-amplitude ratio and optimal configurations is found, with optimal configurations occurring at a ratio of 5.5.

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Nomenclature

A	Amplitude of serrations
C_D	Drag coefficient
C_L	Lift coefficient
C_M	Moment coefficient
$\overline{C}$	Mean chord
D	Drag force
T	Termination distance of serrations
L	Lift force
q_∞	Free-stream dynamic pressure
Re	Reynolds number
S	span
V_∞	Free-stream flow velocity
α	Angle of attack
W	Wavelength of serrations
ρ_∞	Free-stream density
AoA	Angle of attack
CFD	Computational fluid dynamics
DES	Detached eddy simulation
DNS	Direct numerical simulation
SST	Educational wind tunnel
LES	Large eddy simulation
LE	Leading-edge
OC	Optimal configuration
SST	Shear stress transport

1 Introduction

Leading-edge (LE) serrations are biomimetic lift-enhancing devices that are hypothesised to improve the aerodynamic performance of wings through the creation of stream-wise vortex pairs that re-energise the boundary layer and delay flow separation. Quasi-sinusoidal LE serrations are found on the pectoral flippers of Humpback whales (*Megaptera novaeangliae*) seen in Fig. 1 and hypothesised to contribute to their high hydrodynamic manoeuvrability despite their large size and rigid bodies. They were first investigated by Fish and Battle (1995) [10], who observed an increase in lift, especially in the high-angle-of-attack (AoA) regime, a delay in stall, and a reduction in drag. The most common results of the LE-serrations in literature are a softening of the stall profile, replacing the sudden stall onset with a parabolic lift peak best observed in Hansen (2012) [12] or a partial stall onset replacement observed in Johari et al. (2007) [15] with a significant performance improvement post-stall. Immediately post-stall of an unserrated airfoil is where the most improvement is seen with up to a 40% [10, 12, 15] increase in lift. There are notable studies that explore variations in serration properties, such as Lacagnina (2021) [20] and Fernandes et al. (2013) [7]. However, the effect of the variation of serration properties remains unknown. A study by Stanlov and Chong (2019) [31] suggested that the optimal serration amplitude occurs at 2.5% of chord and a wavelength of between 1/10 and 1/5 of chord in their six serration combination study.



Figure 1: Sinusoidal LE tubercles of a humpback whale's (*Megaptera novaeangliae*) flipper.

Increasing manoeuvrability and stall performance on wings and other lift structures has many applications within high-performance aircraft. Most aircraft-specific investigations in LE serrations are conducted on UAVs [3, 34] as UAVs often operate at high angles of attack (AoA) and in highly turbulent environments where the likelihood of stall is great. Wind turbines often operate in off-

design conditions, including yawed flow and unsteady wind conditions where effective AoA is highly variable. The stall-improving measures of LE serrations are highly effective in widening their effective operating range and, therefore, energy production. A similar effect is seen in helicopter rotors where LE serrations have been known to reduce the impact of static and dynamic stall [4], and this extends to applications of fans, drones and other rotational lift devices.

This report will research and identify the gaps in the literature on LE-serrations then produce a computational optimisation using ANSYS Fluent on HPC clusters to determine how the geometric properties of LE-serrations affects aerodynamic performance.

2 Literature Review

2.1 Biological inspiration for LE serrations

Humpback whales (*Megaptera novaeangliae*) have sinusoidal protrusions on the LE of their pectoral flippers known as tubercles seen in Figure. 1. These protrusions are hypothesised to improve hydrodynamic performance and enable high-AoA manoeuvres. First investigated by Fish and Battle [10] in 1995, their unique flipper morphology delays flow separation, delays stall and enhances lift through the generation of streamwise vortices that energise the boundary layer and suppress spanwise flow separation. Miklosovic et al. [24] provided the first experimental validation of these effects by testing serrated leading edges on scaled wing models. Their results demonstrated a stall delay of up to 40%, an increase in the maximum lift coefficient, and a substantial reduction in post-stall drag. These findings sparked the development of biomimetic leading-edge serrations and tubercle-inspired modifications in aerospace and turbomachinery applications, where controlling flow separation and stall is critical.

The typical humpback whale will have 10-11 tubercles on their flipper which explored in Fish et al. (2008) [8] and cover approximately $\sim 70\%$ of the span starting from the tip with typical wavelengths specific to their average chord of $0.35c$ to $0.45c$ ($\sim 1/3c$ to $1/2c$). Explored in Johari et al. (2007) [15], the typical amplitude for the tubercles range from 2.5% to 12% of average chord and Fish et al. (2011) [9] found values of 14.9% to 19.4% specific to local chord. During the lunging manoeuvres the tubercles are hypothesised to aid in, the Reynolds numbers range from 2×10^5 to 1×10^6 depending on the size of the whale with typical speeds of $1 - 4 \text{ m s}^{-1}$ [15].

Another biological source of LE-serrations is that of owls who have fine sawtooth like serrations on their LE but they are much smaller aerodynamic devices than that of humpback whales. Explored in Nafi et al. (2023) [26], their wavelength and amplitudes hover around $1/300c$ with their main affect being a reduction in broadband noise with small $1 - 5\%$ increase in aerodynamic performance metrics depending on specific serration geometry. As the major effect of owl-like serration is aerocoustic, their geometry ranges will be ignored in this study.

2.2 Scope of current LE serration investigations

LE serrations are periodic protrusions that occur on the LE of a wing. In the current literature, many different geometries and study environments have been tested, varying the airfoil, Reynolds number, serration shape/profile, wavelength, amplitude and, minimally, the serration termination distance.

This study will focus on 2D airfoils, where chord, c , and airfoil profile do not change across the span of the wing. The aerodynamics of the relevant studies will be discussed in Section 2.3. This section will define the ranges of the investigated serrations to determine value limits for further investigation as the goal of this study is to attempt to bridge the majority of explored parameter space seen in literature.

2.2.1 Profile/shape

The profile/shape of the serration defines its 2D protrusion profile that varies across studies. Most relevant studies within the current literature investigate a sinusoidal serration profile [2–5, 7, 12, 13, 16–25, 30, 32–35, 37] seen in Figure 2a similar to that of the humpback whale. Sinusoidal serrations are typically denoted as sinusoidal or "wavy" serrations with notable exceptions in Hansen (2012) [12] seen in Figure 2c who explored both horizontal LE variation and also vertical sinusoidal variation denoted as a "wavy" profile seen in Figure. 2c. This is the only notable investigation to study vertically varying serrations.

Triangular profiles seen in Figure 2b are minimally investigated due to their tendency to increase drag [5, 18, 23, 25] to a much greater extent than smooth serrations and their difficulty to mesh to a high-quality in computational studies. A "sawtooth"-like serration profile is commonly investigated in aeroacoustic studies, mimicking the LE serrations seen in Owls as mentioned in Section 2.1. The dimensions of the serrations seen on owls are significantly smaller compared to tubercle studies but, for illustrative purposes, have been scaled in Figure 2d to similar dimensions. No significant aerodynamic papers investigate this profile, but most aeroacoustic investigations control for C_L and C_D and find a minimal aerodynamic effect. Zadorozhna et al. (2021) [36] explored "spherical" or lumped LE-serrations, placing periodic hemispheres along the LE, with aerodynamic improvements at low- Re and low-AoA's. The performance of these serration profiles deteriorates rapidly with increase in flow velocity and AoA and thus Zadorozhna et al. recommended further investigation in the context of small wind-turbines.

This report will restrict itself to the single sinusoidal profile, but as the literature explores non-sinusoidal and irregular shapes, it suggests that variable serration profile is still under-explored.

2.2.2 Wavelength

Variations in the wavelength, W , of the LE serrations have been extensively explored throughout the literature, typically presented as the non-dimensional quantity W/c . The wavelength parameter W is dimensioned in Figure 3. The range of wavelengths investigated in the literature is $0.016/c \leq 1$; however, no study investigates near the full range of values. At the lower end of this range, Ke et al. (2002) [19] investigated five wavelengths in the range of $0.016/c \leq 0.064$ on an S809 airfoil, investigating the performance effect on wind turbines. Lacagnina et al. (2021) investigated three wavelengths in the range of $0.1/c \leq 0.3$ on a NACA 65(12)-10 airfoil, Zhaoyu We et al. (2019) [35] who investigated two wavelengths in the range of $0.25/c \leq 0.5$ on a NACA 634-021 airfoil gave insight into mid-range wavelengths. Manideep et al. (2022) [17] investigated two longer wavelength values in the range of $0.5/c \leq 1$ on a Wortmann FX 60-126 airfoil.

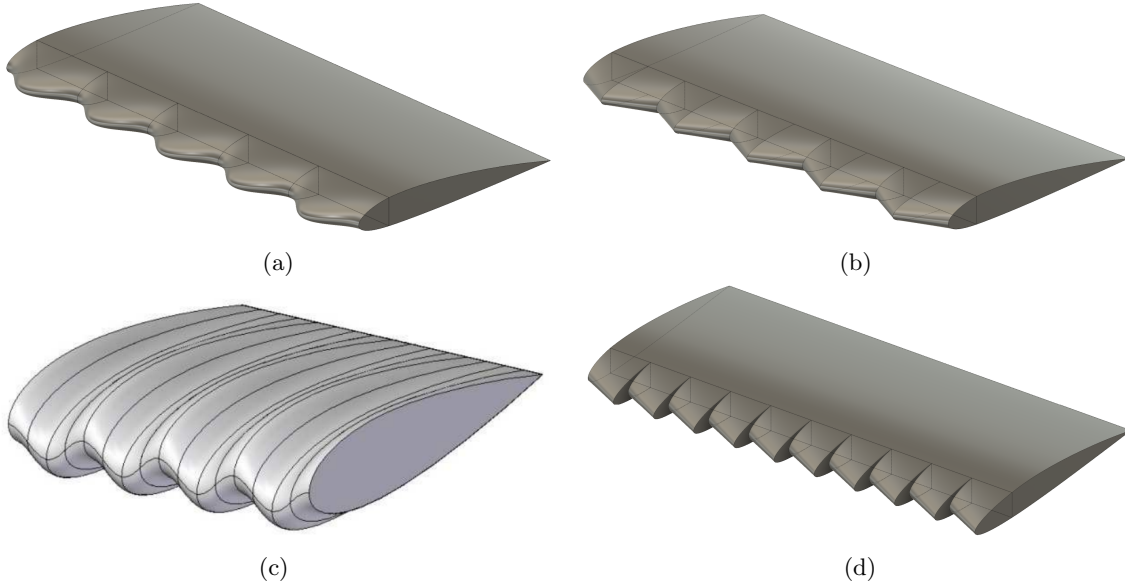


Figure 2: a) Sinusoidal serration profile, b) Triangular serration profile, c) "wavy" sinusoidal serration profile [12], d) "sawtooth"-like serration profile.

In addition to these main wavelength studies, a number of studies have focused on wavelengths chosen to mimic the measured spacing of humpback whale tubercles. Johari et al. (2007) [15] investigated $W/c = 0.25$ and 0.50 on NACA 63(4)-021 wing sections with subsequent finite-wing experiments by Post et al. (2018) [27] who adopted a similar range to study stalled wings. Jayapal Reddy et al. (2022) [14] investigated $W/c = 0.11$, 0.21 and 0.43 on an E216 wind-turbine airfoil at $Re = 10^5$. These bio-mimetic studies dominate the literature and will define the main wavelength parameter sweep conducted.

2.2.3 Amplitude

Variations in the LE serrations' amplitude, A , have also been extensively investigated throughout the literature. Most studies will preserve the planform area of the studied wing across amplitudes by letting the nominal chord length be the position of half amplitude. This can be seen in Figure 3. The amplitude in most studies is presented as the non-dimensional quantity A/c . The range of amplitudes investigated in the literature is $0.01 \leq A/c \leq 0.3$; however, similar to the investigations of wavelengths, no study investigates near the full range of values. Notable comparative studies are done by He et al. (2023) [13] who investigated 4 amplitudes in the range of $0.025 \leq A/c \leq 0.1$ on a NACA 634-021 airfoil, Ke et al.(2002) [19] who investigated 5 amplitudes in range of $0.01 \leq A/c \leq 0.055$ on a S809 airfoil and Stalnov & Chong (2019) [32] who investigated 3 amplitudes in the range of $0.05 \leq A/c \leq 0.5$ on a NACA 65(12)-10 airfoil. The overlap of these studies is difficult to analyse as the serration's performance is highly dependent on study type and geometry. This will be further discussed in Section 2.3.

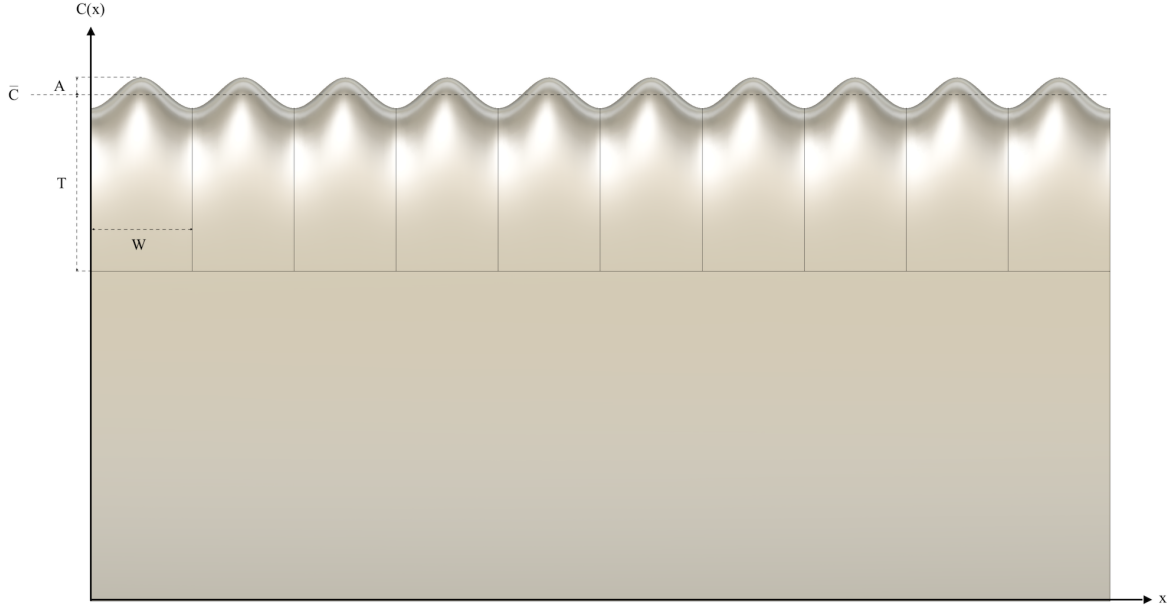


Figure 3: Parameter dimensions of a sinusoidal LE serrated airfoil including A = amplitude, W = wavelength, d = termination distance, c = mean chord, S = span.

2.2.4 Termination distance

The serratation property investigated the least is the termination distance of the serratation, d , seen in Figure 3. No reviewed study varied the termination distance of the serrations, but most studies chose one of two values to be held constant or were unspecified. $1/3c$ and c where the most common termination distances seen in [4, 13, 16, 20, 32, 35] and [2, 3, 5, 7, 17, 19, 30, 37] respectively, with two other studies minimising termination distance and 6 unspecified.

2.2.5 Airfoil

The variation in airfoil profile across studies depends on the wing type a study is investigating. A few of the studies directly investigated and verified the efficacy of serrations mimicking the humpback whale and thus used an airfoil profile similar to that of its flipper. The most common whale-like airfoil used is the cambered NACA 634-021 [13, 34, 35], which mimics the bulbous profile of the whale. Among the studies that investigated the efficacy of serrations on aircraft wings, the most common airfoil profiles are the uncambered NACA 0008 [18], the NACA 0012 [22], the NACA 0015 [21, 25], the NACA 0020 [23, 24] and the NACA 0021 [2, 4, 12] airfoil profiles. General cambered aircraft wing profiles were less explicitly studied, with the only notable study by Fernandes et al. (2013) [7], who investigated the cambered NACA 2412 airfoil profile. A large number of studies investigated rotors where the most common airfoil profile investigated is the NACA 65(12)-10 [20, 32]. Most other investigations used specialised airfoil profiles tailored to their investigation type, namely UAVS, rotors and fans. Only a single study by Hansen (2012) [12] varied the airfoil profile in their investigation, using both NACA 65-021 and NACA 0021 airfoil profiles within minimal aerodynamic difference.

2.2.6 Reynolds Number

Although the range of Reynolds numbers studied was extensive, $2 \times 10^4 \leq Re \leq 2.26 \times 10^6$, the number of studies that varied Re across their serrated wings was limited. The most common Re 's investigated are in the range of $1 \times 10^5 \leq Re \leq 5 \times 10^5$, close to what humpback whales experience during their hydrodynamic maneuvers [24]. The most notable studies found are by Wang et al. 2024 [34] who studied a constant serration geometry across 3 Re in the range of $2.6 \times 10^5 \leq Re \leq 6.3 \times 10^5$ and Lacagnina et al. (2021) [20] who studied a variation in serration geometry across 3 Re in the range of $2 \times 10^5 \leq Re \leq 4 \times 10^5$. Most studies within the literature do not vary the Re within their investigation.

2.2.7 Parametric Studies

Most studies within the current literature do not investigate a wide range of geometric and study properties as the number of required test cases grows rapidly with increasing variation. This is especially relevant for computational studies that require individual simulations for every AoA. Across-study results are difficult to analyse as a lack of variable control decreases comparison efficacy, which is why full parametric studies are essential. The most applicable parametric study within the literature is conducted by Lacagnina et al. (2021) [20] who experimentally studied the complete combinations of $W/c = 0.1, 0.2, 0.3$, $A/c = 0.03, 0.06, 0.12$, and $Re = 2 \times 10^5, 3 \times 10^5, 4 \times 10^5$ on a NACA 65(12)-10 airfoil for a total of 27 test cases. The study used a sinusoidal serration shape and a termination distance of $1/3c$, which were held constant across all test cases. An experimental/numerical parametric investigation was done by Reddy and Sathyabhama (2022) [14], who looked at two low- Re airfoils (E216 and SG6043) with three serration shapes (sinusoidal, slot and triangular) across three amplitudes ($A/c = 0.03, 0.06, 0.11$) and three wavelengths ($W/c = 0.11, 0.21, 0.43$) at $Re = 1 \times 10^5$. This was the largest experimentally tested parametric investigation giving 54 distinct serrated geometries plus baselines. In that study, AoA was varied from 0° to 20° for each configuration, but Reynolds number and termination distance were held fixed.

The most applicable computational parametric study within literature is conducted by Sathyabhama and Sreejith (2022) [30], who computationally studied the complete combinations of $W/c = 0.1, 0.21, 0.41$ and $A/c = 0.013, 0.027, 0.053$ on an E216 airfoil for a total of 9 test cases. The test cases were simulated for 8 AoA values; thus, 72 simulations were run. The study used a sinusoidal serration shape, an unspecified termination distance (most likely c) and a $Re = 1 \times 10^5$, which were constant across all test cases. At higher fidelity, Favier et al. (2012) [6] performed a simple numerical simulation of a NACA 0020 wing at a single AoA at deep stall, varying non-dimensional amplitude $0.01 \leq A/c \leq 0.08$ and wavelength $0.5 \leq W/c \leq 2.0$ of a sinusoidal leading edge. The main result of their investigation is the contour map seen in Figure 4.

Together, these studies demonstrate that full or near-full parametric exploration of $(A/c, W/c)$ is rare, and generally restricted by holding other parameters constant. The results of these studies will be heavily investigated in Section 2.3.

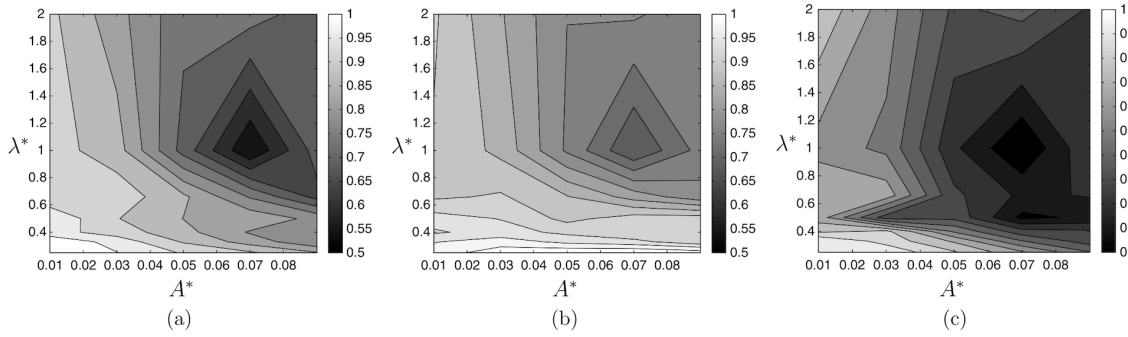


Figure 4: Performance contours of the parametric investigation into serration properties by Favier et al. (2012) [6].

2.2.8 Computational methods

Most applicable studies within the current literature use a combination of CFD and experimental wind tunnel testing. This is primarily due to the inaccuracy of CFD models in predicting stall and post-stall effects.

Ali et al. (2024) [2] investigated the accuracy of certain Reynolds-averaged Navier-Stokes (RANS) CFD turbulence models on LE serrated wings compared to the experimental results found in Hansen et al. (2011) [11]. The accuracy of $k-\omega$ Shear stress transport (SST), $k-\epsilon$ realisable, $(\gamma - Re_\theta)$ SST, Transition $k-l-\omega$ model and Stress- ω Reynolds Stress Model (RSM) models were investigated across 5 AoA's ranging from $0^\circ - 20^\circ$. The model with the highest accuracy across all AoAs is the RSM model, but this comes with a higher computational cost than the other models investigated. The $k-\omega$ SST model was a fantastic predictor of the stall and post-stall region but was limited in predicting the pre-stall region. The worst predictor across all AoAs is the Transition $k-l-\omega$ model. This is corroborated by Lohry et al. (2012) [23] who also concluded that $k-\omega$ SST model was successful in predicting experimental results from Miklosovic et al. (2004) [24].

No applicable study has compared Direct Numerical Simulation (DNS), Large Eddy Simulation (LES) or Detached Eddy Simulations (DES) to the mentioned RANS turbulence models, as these methods require significantly more computational work and are in rarer use for comparative studies. The most commonly used turbulence model within relevant computational studies is the $k-\omega$ SST model [3, 5, 13, 17, 19, 21, 23, 37], which supports the results from Ali et al. (2024) [2]. The $k-\epsilon$ model [7] and $(\gamma - Re_\theta)$ SST [30] are minimally used within literature.

A smaller number of studies have instead employed scale-resolving turbulence models to investigate flow over serrated leading edges. Liu et al. (2022) [22] utilised the LES model and Troll et al. (2021) [25] utilised the DES model since they extensively investigated the vorticity of the flow over LE serrations, of which eddy simulation models are necessary. Pérez-Torró & Kim (2017) [28] carried out a wall-resolved LES of a deep-stalled aerofoil with a sinusoidal leading edge at $Re \approx 1.2 \times 10^5$, who were able to show the counter-rotating vortex pairs due to the high-fidelity simulated eddies. More recently, Ali et al. (2023) [1] used LES to study a serrated NACA 0021 airfoil at $Re \approx 1.2 \times 10^5$ and $\alpha = 20^\circ$ showing the variable performance of serrated airfoils at deep stall. The main draw-back of these LES and DES

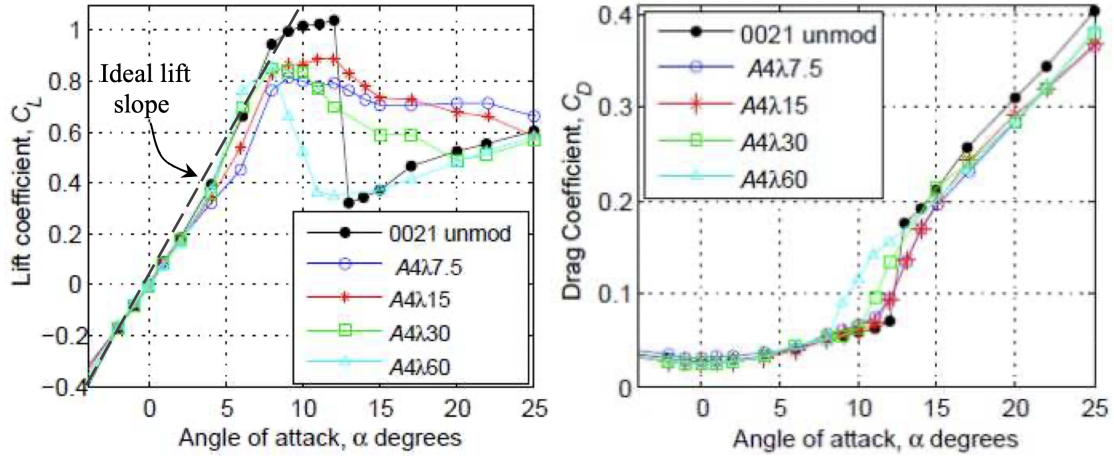


Figure 5: C_L and C_D vs α for various serration configurations seen in Figures 4.17 and 4.18 in Hansen (2012) [12].

studies is the reduced scope due to computational cost restrictions but unambiguously produce the most accurate results.

2.3 Aerodynamic performance

2.3.1 Stall and post-stall characteristics

This section will analyse the results of the optimal configuration (OC) serrations found within the literature. The main aerodynamic change seen within current literature due to LE serrations is the wing's performance in the stall and post-stall regions. Nearly all studies see a dramatic increase in their post-stall lift coefficient, a reduction in stall angle and a "softer" stall profile compared to their baseline airfoil. As seen in Figure 5, the non-serrated lift coefficient curve dramatically drops during stall, but the curves representing the serrated wings exhibit a peak and continuous decrease in lift passing through stall. This is best demonstrated by Hansen (2012) [12], and this result is corroborated by Zhang et al. (2022) [37], Wei et al. (2019) [35] and Zhang et al. (2022) [37] who all denote this as the most significant effect of LE serrations on wings.

This softer stall profile dramatically increases the post-stall lift coefficient. While unanimous across all relevant literature, this effect ranges from a 6.7% [7] increase to a 100% [12] increase in their optimal serration configuration. Most results hover in the range of 14% – 35% [3, 20, 25, 32, 37] but the most significant results come from Borg et al. (2012) [4], Wei et al. (2019) [35] and Hansen (2012) [12] with a 80%, 70% and 100% increase in post-stall lift respectively. This is most likely due to these studies' extensive variable range, the average number of test cases of the three studies is 5, which gave sufficient parametrisation to optimise performance. As seen in Figure 5, the maximum increase in post-stall lift is generally located near the angle of non-serrated airfoil stall and slowly decreases as the angle of attack increases.

Most studies within the current literature report a decrease in the lift coefficient at stall and, thus, a decrease in the global maximum lift coefficient due to the introduction of LE serrations. This result

is seen in the lift profiles in Figure 5. The difference in stall lift coefficients ranges in literature from -17% [4] to a 14% [3] with most results being within -17% to 0% [4, 12, 30, 34, 37]. The greatest increase in the stall lift coefficient within literature is an increase from the baseline airfoil of 40% by Miklosovic et al. (2004) [24]. As this result has not been replicated again in literature and due to the specialised nature of the wing (mimicking a whale flipper), the result is taken as an outlier and thus is inapplicable.

Most studies within the current literature report a change in the stall angle due to the introduction of LE serrations, but the net effect is inconclusive. The literature exhibits a wide range of differences in stall angle from -6° [30] to 10° [3]. The average study reports a net 0° change in stall angle, with most results lying within $\pm 4^\circ$ [4, 7, 12, 20, 32, 35] of this value. The inconclusiveness of the resultant change in stall angle could be due to computational models' inaccuracy in predicting stall, despite both experimental and computational studies within the literature being reviewed.

Most studies within the current literature agree that introducing LE serrations decreases drag in the post-stall region, but this result is variable. In Borg (2012) [4], a 18% reduction in the coefficient of drag is exhibited in their optimal configuration, which is corroborated by Wei et al. (2019) [35] who saw a 25% reduction in post-stall drag and to a lesser degree seen in Hansen (2012) [12], Zhang et al. (2022) [37] and Sudhakar et al. (2017) [33] who all saw an approximate 5% decrease. Most other studies saw a negligible or slight increase in post-stall drag in their optimal configurations [3, 7, 17, 30] but the maximum reported increase in the coefficient of drag within the literature is Manideep et al. (2022) [17] who saw a 25% increase in post-stall drag. This results in a variable but a general increase in lift-to-drag ratio across literature in the post-stall region. The lift-to-drag ratio improvements range wildly from a reduction of 40% seen in Troll et al. (2021) [25], an almost negligible increase seen in Wei et al. (2019) [35] and Wang et al. (2024) [34], to an increase of 300% seen in Sudhakar et al. (2017) [33] in the post-stall region.

The geometries (explored further in Section 2.3.3) and study types explored within this section do not vary significantly, yet the results of each study do. While there are reported trends of aerodynamic change within the post-stall region, the primary outcome of this section is that the results vary wildly, and no single consensus on the aerodynamic effect in the post-stall region can be reached. The lack of consensus within the current literature indicates the need for more research into LE serrations.

2.3.2 Pre-stall characteristics

While some studies have shown minimal pre-stall improvements in aerodynamics, the consensus of the current literature is that there is a slight reduction in performance in the pre-stall regime of a LE serrated wing. As seen in Figure 5, the C_{L_α} and C_{D_α} gradients do not vary, but there is a constant reduction and increase in lift and drag coefficients, respectively, in the pre-stall region. This result is almost unanimously seen within the current literature.

2.3.3 Effects of serration geometry

While extensively studied, the variable effect of serration geometry is still relatively unknown. The whole parameter space of studied serration geometries in the relevant literature is seen in Figure 6. While a cross-study comparison of serration geometry effects is difficult, general trends arise. Most

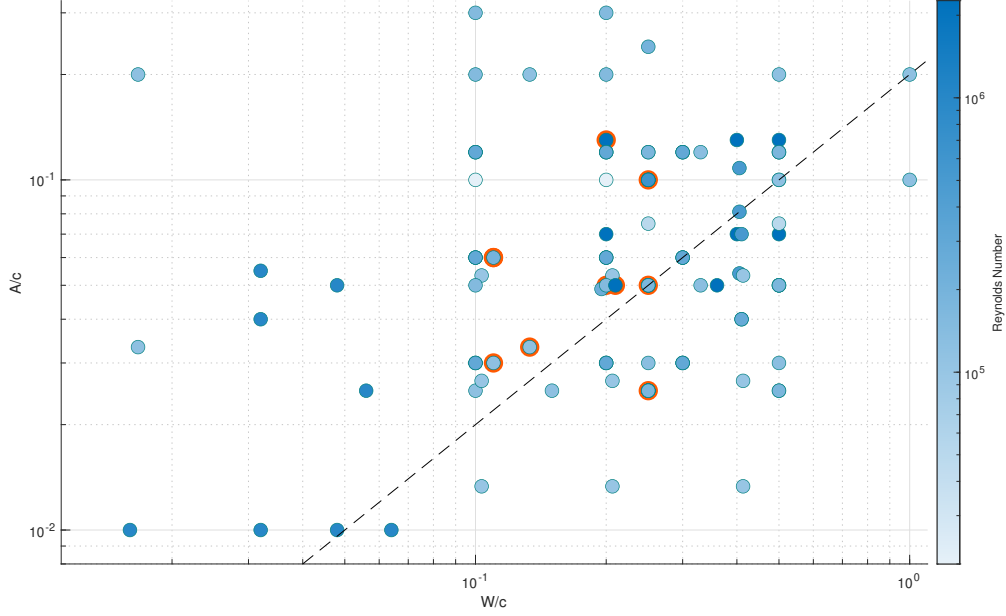


Figure 6: A scatter plot of all the currently researched serration configurations in terms of non-dimensional quantities W/c , A/c and Re with optimal configurations (OC) for each reviewed study marked in red.

studies found that the smaller amplitude and moderate-to-large wavelength serrations performed the best within their studied geometric ranges, with optimal cases usually at the lowest amplitude tested.

The optimal configuration (OC) amplitudes for most studies were $0.025c \leq A/c \leq 0.1c$ with the most common OC amplitude being $0.05c$. Hansen (2012) at OC $A/c = 0.03$ saw a significant decrease in stall angle, a decrease in post-stall lift, and a significant increase in drag around the stall point as the amplitude of the serrations increased, which can be seen in Figure 5. This result is strongly corroborated by Wei et al. (2019) [35], Zhang et al. (2022) [37] and Stanlov and Chang (2019) [32] who denote a sharp lift decrease and drag increase past $A/c > 0.3$. Denoted by Fernandes et al. (2013) [7], the lower amplitude serrations increased maximum post-stall lift but created a faster-decaying lift profile. This result is rare as most studies perceive a constant lift decay as amplitude increases. For most studies, their smallest tested amplitude is their best-performing configuration; thus, the effects of reducing the serration amplitude further are unknown. The pre-stall performance remained relatively constant as amplitude changed, with minor improvements as amplitude decreased.

The wavelengths of the OC for most studies were in the range of $0.15c \leq W/c \leq 0.25c$ with the most common OC wavelength of $0.25c$. Fernandes et al. (2013) [7], Wei et al. (2019) [35] and Zhang et al. (2022) [37] all observed a constant decrease in lift, increase in drag and steeper post-stall lift decays for values of wavelengths that were less than $0.1c$ and greater than $0.35c$. The pre-stall performance remained relatively constant as amplitude changed with minor degradation in performance as wavelengths moved from the optimal range.

Wei et al. (2019) [35] observed that smaller amplitudes combined with smaller wavelengths per-

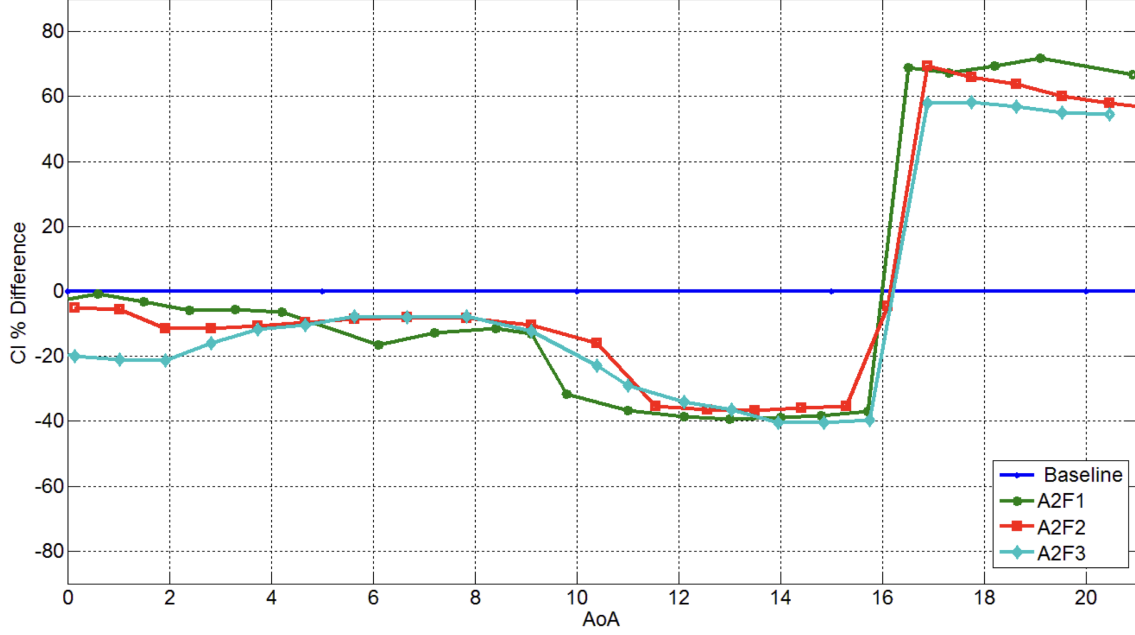


Figure 7: Graph showing the % difference in C_L of the investigated serration geometries to baseline wing in Figure 14 of Borg (2012) [4].

formed optimally. They hypothesised that as amplitudes decrease and wavelengths increase, the serration profile approaches that of an un-serrated wing, which nullifies the effects of the serration. Borg (2012) [4] observed that when a small amplitude is combined with a small wavelength, the increased surface disturbance degrades the flow, hypothesising an optimal balance of properties. The ratio W/A is explored by Lacagnina et al. (2021) [20] who found the optimal ratio between 4 to 6. The line of constant $A/W = 0.2$ ($W/A = 5$) optimised in [20] is plotted in Figure 6 which approximately follows the trend of OCs observed. The existent but weak proportionality indicates other factors must be contributing to their aerodynamic performance, which are not controlled across studies.

No significant correlation of Re , airfoil, serration shape or termination distance is observed in the data collected across or within studies. A single study by Wang et al. (2024) [34] investigated the effect of changing Re , but there is no significant change to the aerodynamic parameters mentioned in this report.

2.3.4 Phenomenological principle

LE serrations alter the flow structures around the wing through targeted vortex generation and flow redistribution. The generated vortices are more stable structures than the general flow, allowing delayed flow separation through re-energising the boundary layer in those regions. Bardera et al. (2024) [3] observed the generation of counter-rotating vortex pairs along the leading edge, which suppresses the generation of laminar separation bubbles, which significantly increases flow attachment and thus improves lift. Laminar separation bubbles form at greater severity at higher AoAs. Therefore,

serrations disproportionally increase the performance of airfoils post-stall. As explained in Manideep et al. (2022) [17], the cause of flow separation is a change in pressure gradient. There comes a point along the wing where this pressure gradient drastically increases, and thus, flow separates. They hypothesised that the vortices generated by the LE serration disturb and "smooth" the drastic pressure increase, which degrades the pressure differential and delays flow separation. Wei et al. (2019) [35] hypothesised that each peak of a serration acts like a highly swept wing LE of a delta wing. This creates a strong streamwise vortex, much like a delta wing's LE vortex (LEV). There are other principles, including the inhibition of the spanwise progression of stall [11] and varying spanwise circulation [29], which are effects due to 3D airfoils which are out of the scope of this report.

2.4 Summary

The relevant literature on the effect of LE serrations is extensive, but there are several gaps in the observed research. The aerodynamic effect of LE serrations in the post-stall regime of a wing seen clearly in Figure 7 are the most promising advantage of their introduction, but the variability of the effects across studies introduces doubt to their overall efficacy. The minimal reduction in performance in the pre-stall region indicates that the serrations would be most efficiently used in high-maneuvring aircraft and not in efficiency-optimised aircraft like passenger planes. The scope of the geometric properties that have been researched is extensive, with the effects of wavelength and amplitude approximately known. The most limited research is in comparative studies across all geometric properties, especially Re and termination distance, where no relevant studies within the current literature were found.

2.4.1 Research Question

Concluding from the literature review summary, the following research question is proposed;

How do the properties of leading-edge serrations affect the aerodynamic performance of wings?

2.4.2 Aims and Objectives

This project will answer the proposed research question by;

1. conducting a parametric computational investigation into the properties of leading-edge serrations (wavelength, amplitude, termination distance) optimising for aerodynamic performance (C_L , C_D , C_L/C_D and C_M) across a range of AoA's and Re 's using computational fluid dynamic simulations,
2. verifying the trends of the computational investigation experimentally using the UNSW Educational Wind Tunnel (EWT).

3 Methodology

To conduct a optimisation at the desired scale, all components of the methodology must be parametrised and automated (or quasi-automated) to reduce the amount of manual input needed. This follows for

all stages of the investigation from the parametric CAD seen in Section 3.1, to the submission of simulation jobs outlined in Section 3.2.2, and the CFD setup itself seen in Section ??.

3.1 Serrated Wing Section Geometry

The sinusoidal serrated wing sections were designed and parametrised in Fusion 360 with variable control over the wing properties of average chord length, $\overline{C}$, and span, S , and serration properties of wavelength, W , amplitude, A , and termination distance, T . All properties are taken specific to the average chord length and are represented as w , a , and t . The variable serration chord profile follows the function;

$$C(x) = \overline{C} - A \cos \frac{2\pi x}{W}, \quad (1)$$

with the termination distance, T , defining the section of the airfoil from the LE that is scaled to match the chord profile defined in Eq. 1 (e.g. at a value of $t = 1$, the entire airfoil profile is scaled to $C(x)$). These profiles ensure that the plan-form area used to calculate aerodynamic parameters remains constant across all models. The wavelengths chosen to be analysed are all integer values of the span to ensure that the wing-section is always symmetric about the centre chord, where the line of symmetry either falls on a trough or a crest. These properties are dimensioned in Fig. 3. The average chord of the model was taken as $\overline{C} = 100$ mm with an aspect-ratio (AR) of two and thus, span, $S = 200$ mm. $AR = 2$ is chosen so that the finer serration combinations can be 3D-printed, surface conditioning is maintained, and the study remains as general as possible. A range of parameter combinations can be seen in Fig. 8. The chosen airfoil profile is the NACA2412 for the broadest application of this study, following from Hansen (2012) [12] who investigated the effectiveness of LE-serrations on several cambered airfoils. The full geometric serration properties explored in this report are;

$$w = W/\overline{C} = \left[\frac{1}{50}, \frac{1}{28.5}, \frac{1}{20}, \frac{1}{13.5}, \frac{1}{10}, \frac{1}{6.5}, \frac{1}{5}, \frac{1}{4}, \frac{1}{3.5}, \frac{1}{3}, \frac{1}{2.5}, \frac{1}{2} \right], \quad (2)$$

$$a = A/\overline{C} = [0.0025, 0.005, 0.01, 0.015, 0.02, 0.025, 0.03, 0.04, 0.05, 0.075, 0.1, 0.15], \quad (3)$$

$$t = T/\overline{C} = [0.1, 0.2, 0.35, 0.5, 0.75, 1.0]. \quad (4)$$

Only the wavelength and amplitude values will be explored entirely at a single initial value of termination at 0.35, which is the most commonly explored termination distance in the literature, second only to fully inserted serrations. Further termination distances will be explored to a lesser extent. Some serration combinations are unrealistic and have been omitted from the study. Incompatible serration combinations occur when the termination distance is smaller than the amplitude. This only occurs for the values of $t = 0.1$ when $a = 0.1$ and 0.15 . Thus the total number of serration combinations is not $12 \times 12 \times = 864$ is reduced to 840. While the complete real combinations of the specific geometric properties is 841 including the unserrated airfoil, the study will focus on 127 real serration combinations at $t = 0.35$ with another 15 used for investigation into termination distance for a total of 142 serrated wing-sections investigated. All 841 serrated wing models were automatically exported using Fusion360's Python API.

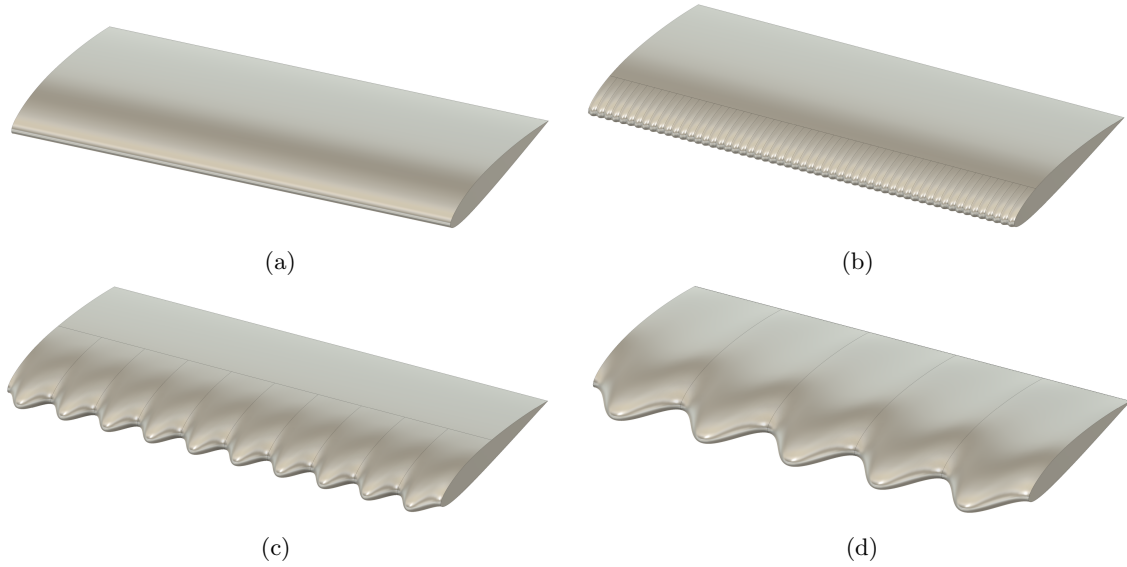


Figure 8: Models with the serration combinations of a) unserrated, b) $w = 1/28.5$, $a = 0.005$, $t = 0.2$, c) $w = 1/5$, $a = 0.05$, $t = 0.5$, d) $w = 1/2.5$, $a = 0.1$, $t = 1.0$.

3.2 Computational Method

3.2.1 Parameter Space

Following Joseph and Sathyabhama (2022) [16] investigation into variable effect of Re on serration performance, an initial, low Reynolds Number value of $Re = 1 \times 10^5$ will be explored with a secondary, high Reynolds Number value of $Re = 5 \times 10^5$. These values encompass the majority of the effective Reynolds numbers in the current literature outlined in Section 2.2.6. The properties of the working fluid will be taken as air at standard sea-level conditions to ensure the best agreement with wind-tunnel testing. With a $\overline{C} = 100$ mm, the corresponding velocities of the low and high Re values are $V_L = 14.61 \text{ m s}^{-1}$ and $V_H = 73.04 \text{ m s}^{-1}$ respectively.

The main result from the parametric investigations detailed in Section 2 is that there are 3 different affected regimes of LE-serrations, namely, how they perform pre-stall, at stall, and post-stall. To effectively study the three regimes of interest without resolving the complete aerodynamic profiles (which could need > 20 AoA's), initially, the stall profiles of the unserrated airfoil and selected serration combinations will be fully simulated to determine the necessary 3 AoA's to describe the pre-stall, stall and post-stall regimes at both Re . An additional 3 AoA will be simulated when investigating the termination distance.

For 3 AoA values, at 2 Re values, across 127 models, this results in a sub-total of 762 simulations for the amplitude and wavelength investigation. For the investigation into termination distance, 6 AoA values, at 2 Re values, across 15 models results in a sub-total of 180 simulations. Thus, in total for the entire data collection, 942 simulations will be run. This does not include the needed simulations for R&D and CFD setup determination.

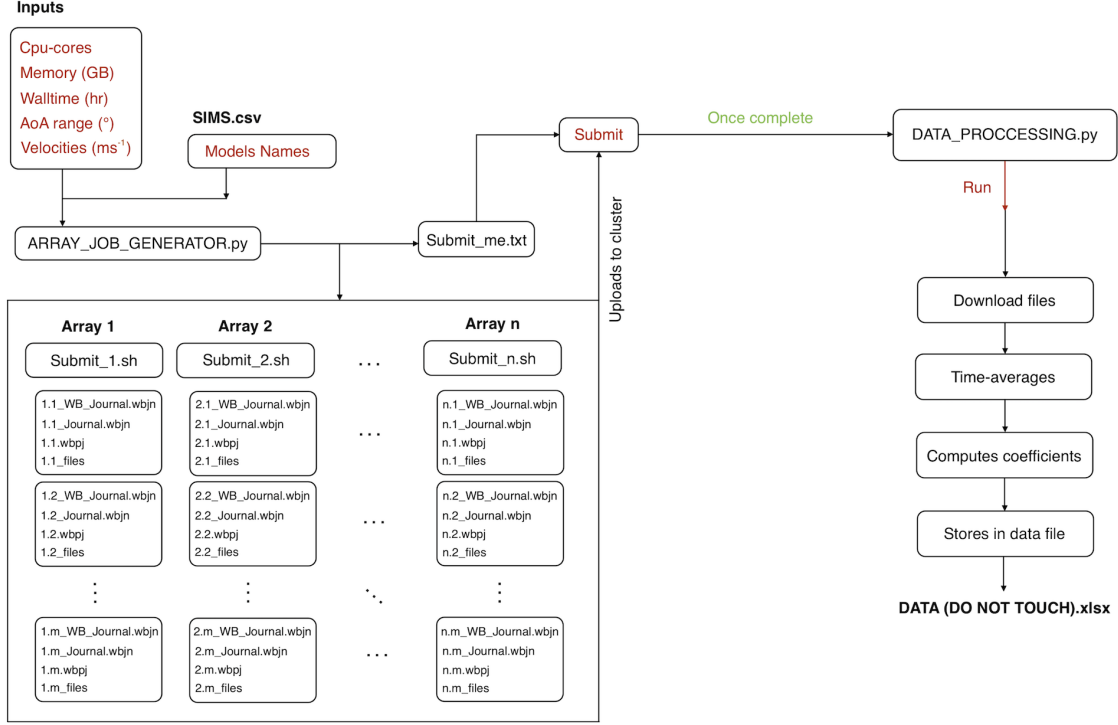


Figure 9: Flow diagram of the automation workflow for the simulation with the manual inputs and tasks denoted in red.

3.2.2 HPC Automation Workflow

To be able to simulate the complete combinations of the serration properties detailed in Section 3.2.1, the simulations will be run on two available clusters, KATANA at UNSW, a 7,600 cpu-core cluster, and GADI at NCI, a 260,000 cpu-core cluster. Supplied through UNSW ResTech, KATANA has no allocated computational usage limitations but has a simultaneous core usage limit of 576 on the queue which equates to 12x48-core simulations or 9x64-core simulations.

In contrast, access to GADI is through service units (SU) allocation (1 SU = 2 core-hour) supplied through Con Doolan’s project allocated by UNSW. The project fk00 has multiple members which share the allocation of 1100 kSU’s per term which cannot be fully utilised through this investigation (as others need allocated SUs). The queue on GADI has a simultaneous limit of 100 individual jobs (no core usgae limitations) which makes GADI the significantly more powerful cluster but is purely limited by the total computational work. The majority of the R&D is completed on KATANA and the final data collection is conducted on GADI. Both clusters have the ability to run array jobs which will be utilised to run multiple AoA for each model.

In order to reduce the manual input it takes to run 942 individual simulations, a automation workflow was setup on a local system that interfaces with the Simple Self Host (SSH) and the File Transfer Protocol (FTP) capabilities of both clusters. This workflow is seen graphically in Figure 9. The automation paths are scripted in Python. The initial step is to define the input parameters. The desired

model names are input into a .csv and the setup parameters of AoA range, velocity values, desired cluster, walltime, memory and cpu-cores are input into the script ARRAY_JOB_GENERATOR.py. Using these inputs, the script generates a single submit script in BASH per model which sets up an array job. A array job allows for the single submit script but can run many simulations per model (runs all AoA and Re automatically per model). For each job in the array, the script generates an ANSYS Workbench Journal script, the ANSYS Fluent Journal script and copies the BASE_SIM (expanded further in Section 3.2.4). After all arrays have been created, the script then uploads the arrays to the specified cluster (which is the longest part of the process as summed file sizes were of $O(GB)$. The BASE_SIM was purposely kept as low of a file as possible to reduce this) and a SUBMIT_ME.txt is created. Contained within SUBMIT_ME.txt is the automatically generated submit command for all the array jobs. This step was kept manual to accommodate for errors and variability in the desired subset of the generated array jobs to submit. The cluster is logged into on the BASH terminal PuTTY and the submit command is paste in. This submits all the array jobs at once which can be monitored on the user queue. This automation workflow (which was the longest device to develop for this entire thesis) reduced the manual user input drastically.

After partial completion or completion of the array jobs, the script DATA_PROCESSING.py is run to download the generated data, time-average the force values, generate the necessary coefficients and save the data in a central data csv. The output for each simulation is time-varying aerodynamic property files which are scrapped from both cluster and downloaded maintaining simulation history in the naming convention.

3.2.3 Fluid Domain and Boundary Conditions

The computational domain was chosen according to three conditions, 1. to minimise boundary feedback across the wing-section, 2. remain representative of the Educational Wing Tunnel (EWT) and 3. keeping the computational cost of the simulations reasonable.

The final fluid domain relative to the position of the quarter chord centre of the wing-section is;

$$\pm 5C \text{ in the vertical direction, } 4C \text{ upstream, } 12C \text{ downstream, } \pm 2.5C \text{ spanwise,}$$

where $\overline{C} = 100 \text{ mm}$ and thus the fluid domain has absolute dimensions of $1600 \text{ mm} \times 1000 \text{ mm} \times 500 \text{ mm}$. The domain dimensions were selected through a set of preliminary simulations in which the upstream, downstream, vertical, and spanwise dimensions were increased until the differences in lift and drag values at pre-stall and post-stall AoAs were small. Four domain sizes were considered, a small domain representative of the EWT test section of $10\overline{C} \times 3\overline{C} \times 3\overline{C}$, medium domain (the final dimensions), large domain of $25\overline{C} \times 10\overline{C} \times 10\overline{C}$ and pressure far field condition at the large domain size. Three models were analysed both pre-stall and post-stall at $Re = 10^5$ (simulations were conducted using the final CFD model and setup (final setup seen in Section 3.2.6) process was iterative). The results of the independence study is in Table 1;

While complete convergence was not reached, the domain was not increased further to reduce computational cost of the simulations. An additional consideration was consistency with the EWT test section. The spanwise width was kept of the same order as the physical tunnel to maintain similar

Table 1: Results of domain independence study at $Re = 10^5$.

SC	AoA	$(C_L)_{\text{small}}$	$(C_L)_{\text{medium}}$	$(C_L)_{\text{large}}$	$(C_L)_{\text{large f.f.}}$
$[0, 0, 0]_{SC}$	16	0.8522	0.8314	0.8307	0.8304
	25	0.6464	0.6231	0.6232	0.6224
$[1/5, 0.075, 0.35]_{SC}$	16	0.7918	0.7725	0.7711	0.7713
	25	0.7661	0.7534	0.7530	0.7541
$[1/13.5, 0.005, 0.35]_{SC}$	16	0.8571	0.8269	0.8268	0.8259
	25	0.7209	0.6923	0.6906	0.6915

blockage and wall-induced effects between the computational and experiment tests. Similarly, while utilising periodic boundary conditions on a sing-section would eliminate the end effects and top vortices produced, it would not be representative of experimental data and the computational results could not be verified. The complete set of boundary conditions used, labelled in Figure 10, are;

- **Inlet:** A velocity inlet was used on the upstream plane to create the desired free-stream velocity, U_∞ corresponding with the Re values specified in 3.2.1.
- **Outlet:** A pressure outlet boundary condition was used at the downstream plane, with a gauge pressure of zero.
- **Farfield:** The top, bottom, and spanwise side boundary planes of the domain were treated as slip walls. This approximates far-field pressure conditions and thus unbounded flow while keeping a computational affordable domain and avoiding convergence issues.
- **Bodies:** All aerodynamic surfaces were treated as no-slip walls.

While the model and flow conditions are symmetric, a symmetric boundary condition needs the extra condition of symmetric fluid flow about the plane. Symmetric boundary conditions do not allow flow normal to the surface and would thus fail in capturing the spanwise flow phenomena of vortices. The reduction of computational cost by two is not worth this reduction in accuracy.

The AoA of the wing was set as a rotation about quarter chord of the wing-section and not that of a rotation of the freestream velocity direction. One could change the inlet to be cylindrically symmetric and just change the flow direction to avoid remeshing every simulation but in order to capture shedding phenomena in the wake region, the wake region must have a fine mesh. If this wake region is not fixed in space, this region would have to be much larger to accommodate the delta in angle which would waste computational work. The rotation about quarter chord was set as the input parameter “AoA” and would be varied automatically in the study.

3.2.4 Meshing

The goal of the ANSYS Fluent CFD setup in determination is to create a BASE.SIM as detailed in Section 3.2.2 that produces accurate results across all models and simulation environments. The three variables that change across simulations is the model, the AoA and the fluid velocity. The change in model, specifically, how fine and intense the serrations are, will change how high-fidelity the surface

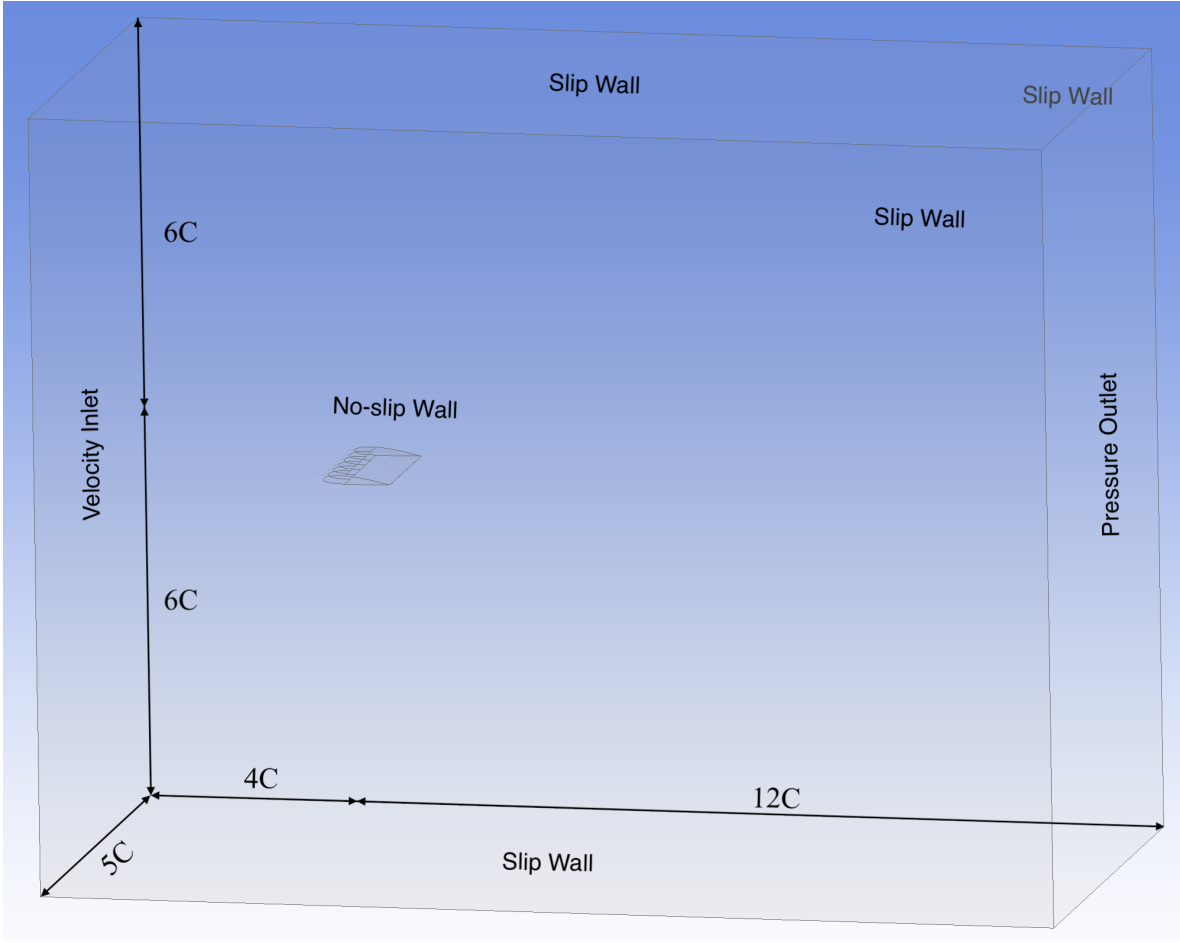


Figure 10: Dimensioned fluid domain showing boundary conditions.

mesh is around the serrations to capture its geometry successfully. This results in the need to create meshing conditions that are variable to the geometry of the serrations. A hybrid unstructured mesh was used, with a 30-layer first height boundary layer mesh on the wing surfaces with a predominantly poly-hexcore volume mesh in the greater fluid domain.

In ANSYS Fluent Meshing, to create this variable mesh fidelity on the wing surface, curvature and proximity surface local sizing are used. They both take in a minimum and maximum cell size with the variable condition being “Curvature Normal Angle” and “Cells per gap” respectively. The curvature normal angle condition specifies the maximum allowable delta in the surface normal vector between adjacent mesh elements. When the curvature causes the surface normal to rotate by more than a threshold angle, the mesher reduces the element size to resolve the feature. This condition is perfect for the curved nature of the sinusoidal serrations investigated in this report. The cells per gap defines the minimum number of cells that must span the distance between two opposing surfaces.

In this study, the curvature normal angle on the serrated leading edge was set to 7° which ensured the sinusoidal profile of each serration was resolved smoothly. The minimum cell size on the surface of the serrations was set to 0.05 mm with maximum of 0.5 mm thus the smallest wavelength serration

of 2 mm at a wavelength of $w = 1/50$ has 40 cells across its length. Along with a proximity local sizing condition on the body and size of the wing-section with the same minimum and maximum local cell sizes, across all geometries this produced surface meshes with roughly 200,000 – 500,000 surface mesh cells on the wing. These values were determined through a mesh independence study conducted on three fidelity levels, a course-fidelity mesh at $[\min, \max] = [0.2, 1.5]$ mm which allows for 10 cells across the minimum wavelength, a medium-fidelity mesh at $[\min, \max] = [0.05, 0.5]$ mm which allows for 40 cells across the minimum wavelength (final values) and a fine-fidelity mesh at $[\min, \max] = [0.2, 0.25]$ mm which allows for 100 cells across the minimum wavelength. The volume mesh sizes for each simulation ran respectively were in range of $[1 \times 10^6, 4 \times 10^6]_{\text{course}}$, $[4 \times 10^6, 11 \times 10^6]_{\text{med}}$ and $[14 \times 10^6, 31 \times 10^6]_{\text{fine}}$. Three models were analysed both pre-stall and post-stall at $Re = 10^5$ (simulations were conducted using the final CFD model and setup (final setup seen in Section 3.2.6) process was iterative). The results for the mesh independence study is in Table 3;

Table 2: Results of mesh independence study at $Re = 10^5$.

SC	AoA	$(C_L)_{\text{course}}$	$(C_L)_{\text{medium}}$	$(C_L)_{\text{fine}}$
$[0, 0, 0]_{SC}$	16	0.8401	0.8314	0.8311
	25	0.7100	0.6231	0.6229
$[1/5, 0.075, 0.35]_{SC}$	16	0.7798	0.7725	0.7717
	25	0.7992	0.7534	0.7501
$[1/13.5, 0.005, 0.35]_{SC}$	16	0.8356	0.8269	0.8277
	25	0.7416	0.6923	0.6951

The course mesh over estimates the lift coefficient especially in the post-stall AoA which makes sense as a courser mesh would not be able to accurately capture the flow separation phenomena needed for accurate stall prediction. The values for C_L change minimally in the fine mesh compared with the medium mesh with a 3-4 times increase in final mesh size and thus the medium surface mesh size is chosen. The final surface mesh fidelity is seen in Figure 11.

To capture the wake region of the wing-section, a mesh refinement region was defined around the wing. This is especially important for Large Eddy Simulation (LES) turbulence models used in this report (justification found in Section 3.2.5) as the region must have high enough fidelity to capture the eddy structures. The perturbed region of the flow around the wing was determined by looking at pressure contours about the centre plane of the wing-section. The maximum region around the wing for any AoA was $0.3\bar{C}$ upstream, ± 0.95 vertically, $\pm 1.3\bar{C}$ spanwise and $11\bar{C}$ downstream (wake lasted to the end of the fluid domain). The majority of the eddies observed in the flow in a high-fidelity test run ended at $4\bar{C}$ downstream and thus the region for which a fine mesh refinement is applied was determined to be $\pm 2\bar{C}$ vertically, $0.5\bar{C}$ upstream, $6\bar{C}$ downstream and $\pm \bar{C}$ spanwise. This region is defined past the boundary of the model and thus changes with AoA and geometry which is perfect for the variable conditions of the computational study. To determine the necessary cell refinement size a wake region mesh refinement study was conducted at 3 mesh sizes, course-mesh at 3 mm, 6 times larger than the largest surface mesh cell size and $0.03\bar{C}$, medium-mesh at 1 mm, 2 times larger than the largest surface mesh cell size and $0.01\bar{C}$, and a fine-mesh at 0.5 mm, the same fidelity as the largest surface mesh size and $0.005\bar{C}$. The volume mesh sizes for each simulation ran respectively were in

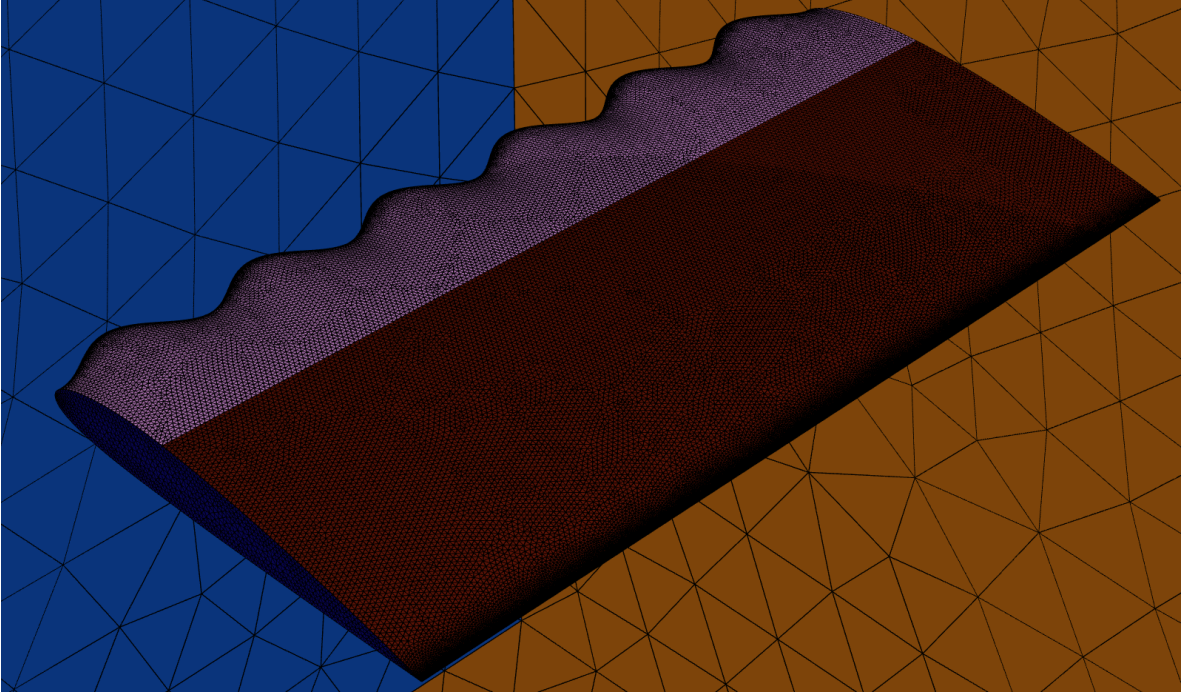


Figure 11: Image of the surface mesh fidelity on the $[1/3, 0.04, 0.35]_{SC}$ model.

range of $[0.5 \times 10^6, 3 \times 10^6]_{\text{course}}$, $[4 \times 10^6, 11 \times 10^6]_{\text{med}}$ and $[9 \times 10^6, 22 \times 10^6]_{\text{fine}}$. Three models were analysed both pre-stall and post-stall at $Re = 10^5$ (simulations were conducted using the final CFD model and setup (final setup seen in Section 3.2.6) process was iterative). The results for the wake region mesh independence study is in Table 3;

Table 3: Results of wake region mesh independence study at $Re = 10^5$.

SC	AoA	$(C_L)_{\text{course}}$	$(C_L)_{\text{medium}}$	$(C_L)_{\text{fine}}$
$[0, 0, 0]_{SC}$	16	0.8373	0.8314	0.8334
	25	0.6341	0.6231	0.6201
$[1/5, 0.075, 0.35]_{SC}$	16	0.7619	0.7725	0.7736
	25	0.7289	0.7534	0.7581
$[1/13.5, 0.005, 0.35]_{SC}$	16	0.8112	0.8269	0.8289
	25	0.6707	0.6923	0.6929

The course-mesh wake region massively under predicts the post-stall regime values of lift which is intuitive as the lower-fidelity mesh would not as effectively model the stable low pressure structures that result in improved performance due to LE-serrations. This improves dramatically into the medium fidelity mesh and only increase slightly, $< 0.5\%$ when using the high-fidelity wake region. The medium mesh refinement value of 1 mm, 2 times larger than the largest surface mesh cell size and $0.01\bar{C}$ is used.

The last mesh fidelity consideration is the boundary layer that is formed along the wing-section's surface. For LES models, the viscous sub-layer must be resolved and following from Liu et al. (2022)

[22], the value of $y^+ < 1$ is necessary. y^+ is found by;

$$y^+ = \frac{yu_\tau}{\nu}, \quad (5)$$

where y = first layer height, u_τ = wall shear stress, and ν = dynamic viscosity. The wall shear stress is proportional to the flow velocity over the serrated wing-section and thus will have to be variable across simulations to remain accurate. The best way to estimate the y^+ of simulations is to run an initial guess and iterate until a value of < 1 is found. For $Re = 10^5$ with $V_L = 14.61 \text{ m s}^{-1}$, the first layer height which produced a suitable y^+ is $y = 0.005 \text{ mm}$ and for $Re = 5 \times 10^5$ with $V_L = 73.04 \text{ m s}^{-1}$, the first layer height which produced a suitable y^+ is $y = 0.001 \text{ mm}$ (5 times as less as the low- Re which is expected). The thickness of the boundary layer was estimated visually after simulating a range of serration combinations with a maximum value of 4.72 mm. Following Liu et al. (2022) [22] who determined a 30-layer boundary layer was suitable for LES over serrated airfoils, the growth rate is calculated to be 0.212.

3.2.5 CFD Model

The aerodynamic behaviour of the air around the serrated wing-sections spans separated flow, partially separated and deeply separated at various angles of attack. The capturing of these phenomena is a difficult process that needs a strong turbulence model. The best turbulence model was not determined through a purposful convergence study but was found organically through trial and error through trying to recreate past results. The process was the following;

- Steady RANS: steady-state $k\omega$ -SST was used in the early stages until high AoA regimes were explored and convergence issues were faced,
- Unsteady RANS (URANS): transient $k\omega$ -SST were then performed which converged but failed to recreate the data of previous studies,
- Large-Eddy Simulation (LES): LES was used which matched previous results to a high degree and was the final turbulence model used in the parametric optimisation.

The initial steady-state simulations would match previous results well for low, pre-stall AoA but would fail to converge at higher regimes. The model attempts to force steady mean flow in a regime that is strongly unsteady, leading to convergence difficulties with non-physical behaviour like oscillatory residuals and in some cases divergence. Moving to transient CFD in unsteady URANS, the convergence issues dropped and would converge even past 30° . The issue was that the stall profiles did not mimic that of previous experimental data. Seen in Figure 12, the URANS serrated simulation massively over-predicted stall, stalling 5° sooner than that of the unserrated airfoil and minimal benefits post-stall. These results were unanimous across many serration combinations, none of them matching any experimental data in the literature. After investigating Liu et al. (2022) [22] study into using LES on serrated airfoils, the choice to move to LES was made. The comparable data for LES is seen in Figure 12 compared to data made in Hansen (2012) [12] which matched to a much greater degree and saw the benefits of LE-serrations in the post-stall regime.

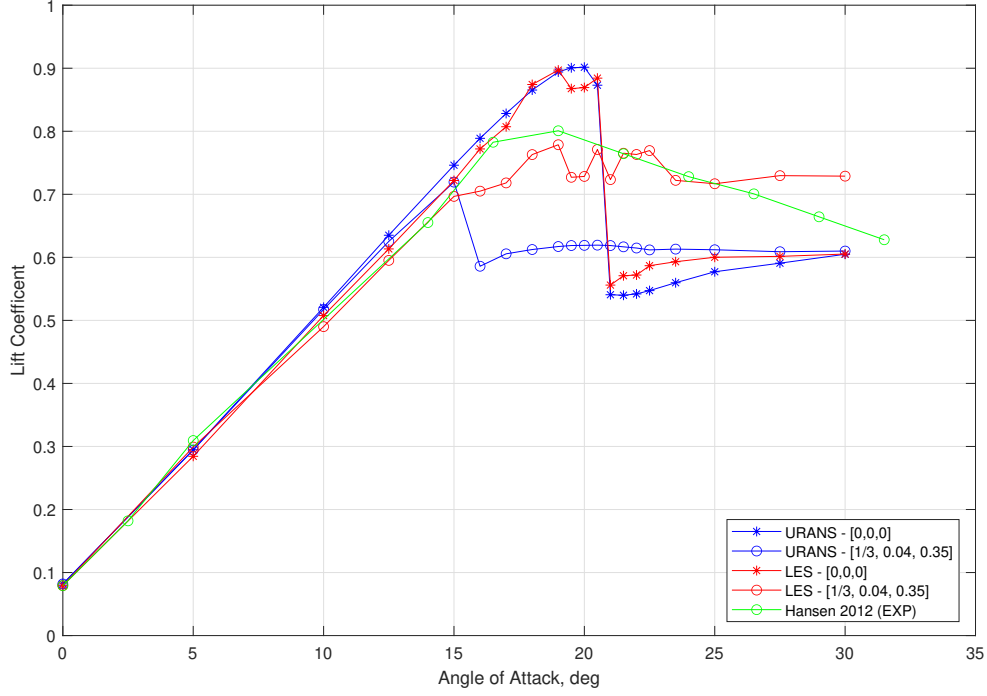


Figure 12: Plot of the URANS and LES data for the unserrated airfoil and the $[1/3, 0.04, 0.35]_{SC}$ compared to data from Hansen (2012) [12] seen in A4λ15 in Figure 5.

In LES, the larger energy-containing turbulent eddies are resolved directly while only the subgrid-scale (SGS) motions are modelled. This is particularly advantageous for LE-serrations where vortical structures generated at the serrations play a central role in the aerodynamics. The LES was performed on the refined meshes described in Section 3.2.4. Following the setup seen in Liu et al. (2022) [22] and Pérez-Torró & Kim (2017) [28] LES study on deep-stalled LE-serrations, the necessary Courant–Friedrichs–Lewy (CFL) number, defined as;

$$CFL = \frac{u\Delta t}{\Delta x}, \quad (6)$$

where u = local velocity, Δt = time-step and Δx = cell size, for LES in near wall phenomena needs to hover around $1 - 2$ to accurately describe the sub-eddy turbulence. At the scope of the study detailed in Section 3.2.1, completing all simulations with the equivalent time-step of $CFL = 1 - 2$ would be intensely computationally expensive. A compromising, two step process was formed to reduce the computational work while keeping accuracy. The process is;

- 1. Run with a $CFL = 10 - 50$ for 200 time-steps to push past initial transients and stall,
- 2. Reduce down to a $CFL = 1 - 5$ for an additional 1000 time-steps to capture the sub-eddy phenomena.

The time-varying data is then average in the last 200 time-steps and the coefficients are calculated and

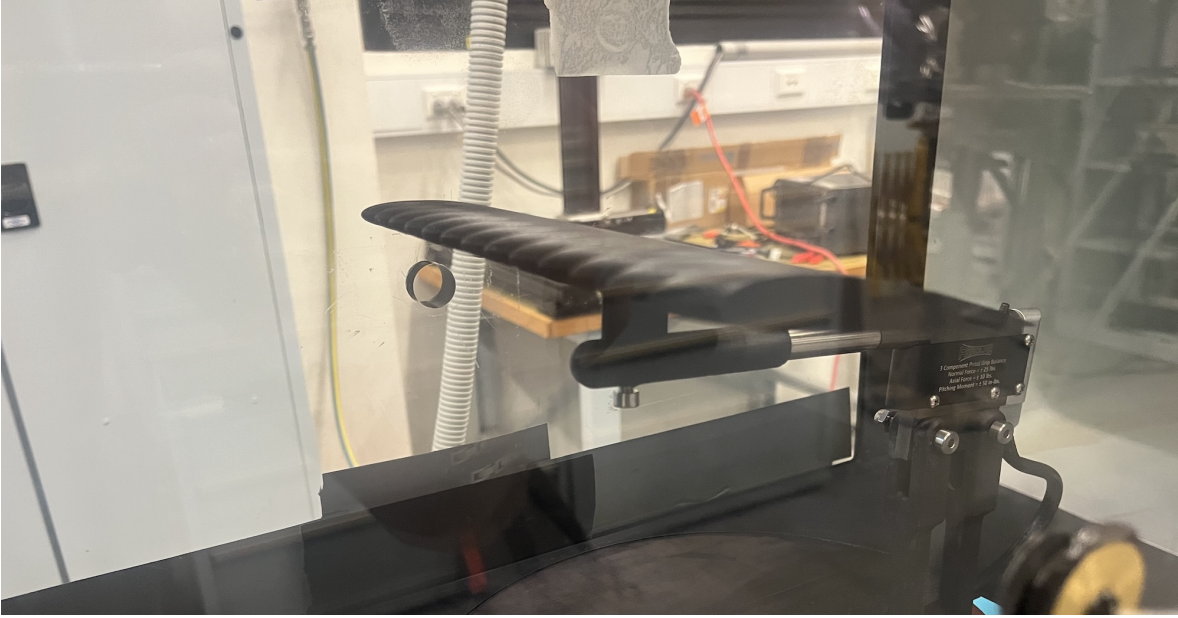


Figure 13: Experimental wind-tunnel setup with stand-off mount in the EWT.

plotted. The first stage is done through ANSYS Workbench scripting, where the model is meshed, and solved. The data is saved to a output file where it is subsequently opened by ANSYS Fluent scripting where the CFL is reduced and solved again. The data, as described in Section 3.2.2, is automatically scraped from the cluster and processed.

3.2.6 Computational Model Summary

The computational model used for the parametric optimisation is computational intensive but has been refined to still capture the transient phenomena and be accurate to historical and this studies own experimental data. Following from the mesh and domain determination in Sections 3.2.3 and 3.2.4, the final volume cell number ranged from 4,000,000 to 15,000,000 cells. With the time-step formulation described in Section 3.2.5, each simulation took approximately 4-12 hours to solve on either clusters 48 cpu-core nodes. In total, 942 simulations were run for data collection, and the total computational work for both collected data and R&D was 550,000 core-hours across both clusters.

As the settings for each simulation are optimised to be accurate across a range of models rather than specific to a single model, the computational results in this report are presented as trends rather than explicitly accurate values.

3.3 Experimental Method

The experimental setup was simple and was intended only to verify the accuracy of the trends in the computational results. The dimensions of the EWT seen in Fig. 13 are 30.48 cm $\times$ 30.48 cm $\times$ 60.96 cm equipped with a 3-component sting balance with max AoA of 23°. As the AoA range is greater than this value, a mount was designed to offset the model by 30 mm (this also reduces sting balance interference

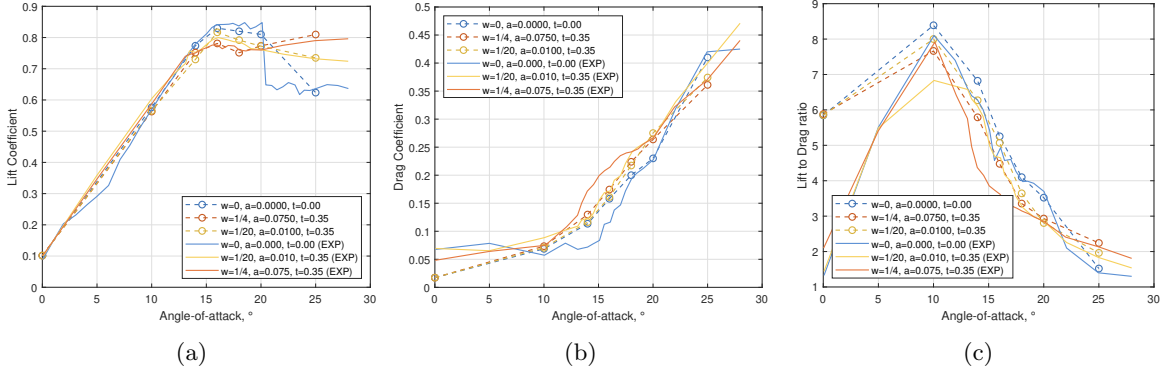


Figure 14: Stall profiles comparing experimental data and computational data for three models across the aerodynamic properties a) Lift Coefficient, b) Drag Coefficient, and c) Lift to Drag ratio.

with the flow), which allowed for a -5° offset to be designed into the mount. This expanded the maximum AoA to 28° , which sufficiently contains the computational domain.

The tests were conducted at a freestream velocity of $V_L = 14.61 \text{ m s}^{-1}$ corresponding to the chord base $Re = 1 \times 10^5$. The frontal blockage ratio was below 2.5% even for the greater AoA, so no blockage correction was applied. The models were 3D-printed from PLA, with the print "grain" constant in the stream-wise direction across all models. The AoA step was large at the edges of the sweep, with $\Delta 0.5^\circ$ steps over the centre of the sweep at stall. The data acquisition (DAQ) was set to a 10 sec multi-point collection at 100 data points a second and the force coefficients were calculated from the 1000 point average. The sting balance records normal and axial forces which were transformed to lift and drag values taking into consideration the effective AoA delta given by the geometry of the mount.

Three models were tested, $[0, 0, 0]_{SC}$, $[1/4, 0.075, 0.35]_{SC}$, and $[1/20, 0.01, 0.35]_{SC}$, the unserrated airfoil and a low-intensity serration combination and high-intensity serration combination.

4 Results and discussion

4.1 Experimental Verification

To verify the accuracy of the computational setup and to determine the necessary AoAs to simulate, a larger AoA sweep was conducted experimentally and computationally across an unserrated and a low- and high-fidelity serration combination. The results are compared in Figure 14.

The lift profiles in Figure 14a captured by both the computational and experimental measurements are highly comparable, with nearly all phenomenological differences in model profiles captured for both. Whereas the exact stall angle is not determined computationally, the values for the coefficient of lift deviate a maximum of 5% for any of the models. This result is surprisingly accurate. The differences increase with the values for drag in Figure 14b. The drag across all models is under-estimated by over 50% at $\text{AoA} = 0^\circ$. However, for both serrated airfoils, the values return to the experimental trends, with mild deviations in the lower-fidelity serration around 15° . Past this value, the drag values for both the experimental and the computational results closely agree in the post-stall regime. The unserrated airfoil computational values deviate significantly until approximately 16° . This pre-stall

inaccuracy is surprising and is best seen in Figure 14c, where the computational values overestimate the experimental data by up to 6 times. This is most likely due to the model’s surface roughness, which is not computationally simulated. For the sake of accuracy, the pre- 15° AoA regime will not be used in the parametric optimisation, as accuracy cannot be guaranteed. Otherwise, the closeness of both the computational and experimental data seen past 15° gives confidence to the accuracy of a parametric optimisation for AoA = 16° , 20° , and 25° which represent the pre-stall, stall, and post-stall performance of the serrated wing-sections respectively and will give enough fidelity to see performance trends in all three regimes.

4.2 Low Reynolds Number

The low Re investigation will be conducted at $Re = 1 \times 10^5$ with a fluid velocity of $V_L = 14.61 \text{ m s}^{-1}$ as described in Section 3.2.1.

4.2.1 Wavelength and Amplitude

The value of termination distance, t , will be held constant at the value 0.35 for the initial optimisation, and the variable effect of termination distance will be explored in Section 4.2.2. For succinct analysis, a specific serration combination (SC) will be referenced in the format of $[w, a, t]_{SC}$ in this section. As stated in Section. 4.1, the parametric optimisation will be run across AoA = 16° , 20° , and 25° which represent the performance of the wing-sections in the pre-stall, stall, and post-stall regimes respectively. Seen in Figures 15, 16, 17, and 18, the serration intensity increases as you move from the bottom right-hand to the top-left hand corner of the plots following the increase in the parameter w/a . The black sections of these plots are the serration combinations deemed unrealistic and not simulated (eliminated from the solution space based on inspection, significant reductions in aerodynamic trends, and mesh size/quality). The performance coefficients are plotted against the geometric parameter w/a to observe the relationship between w/a and its performance in Figures 19, 20, 21, and 22.

The pre-stall trends at 16° observed in Figure 15a, 16a, and 17a are extremely clear with a wide “island” of well performing serration combinations following the line between $[1/50, 0.0025, 0.35]_{SC}$ and $[1/4, 0.04, 0.35]_{SC}$ with the approximate gradient w/a falling between 5 to 30 as observed in Figure 19 with peak performance at a $w/a = 10$. These results are loosely corroborated by Lacagnina et al. (2021) [20] who found the optimal w/a ratio to be between 4 and 8 in their experimental parametric study. Seen in Figure 15a, the high-lift “island” is wide and contains the majority of the serration combinations, which indicates that pre-stall lift is relatively insensitive to serration geometry. This performance is only given if w/a does not fall far from the aforementioned values of $w/a = 8 - 12$ and the serration amplitude is not too large. The dominance of performance in the low-amplitude range indicates that, in pre-stall regimes, small-amplitude and wavelength combinations are favoured. The observed “island” of low drag serration combinations in Figure 16a is noticeably thinner than that of lift. The w/a drag distribution is shown in Figure 16a and is semi-bimodal, with the lowest drag values again falling around $w/a = 10$ and a second trough at very low w/a . This suggest that there are two methods of drag reduction within the parameter space at different values of w/a which will be explored later. The effect of the stark increase in drag across very-low- and very-high-fidelity serrations is best seen in the values for the lift-to-drag ratio in Figure 17a. The increased drag at low-fidelity

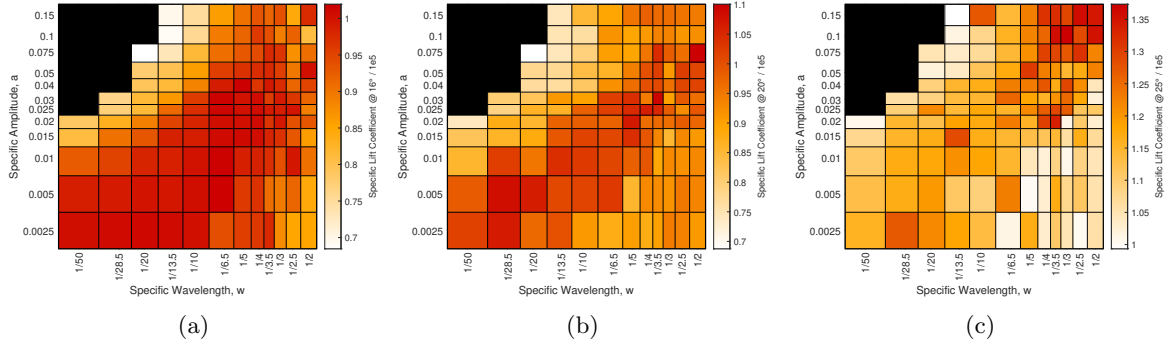


Figure 15: Log-log plots at $Re = 100,000$ of the parameter combinations of specific wavelength, w , and specific amplitude, a showing the specific aerodynamic property C_L/C_{L_0} for a) pre-stall regime of $AoA = 16^\circ$ for $C_{L_0} = 0.83$, b) stall regime of $AoA = 20^\circ$ for $C_{L_0} = 0.81$, and c) post-stall regime of $AoA = 25^\circ$ for $C_{L_0} = 0.62$.

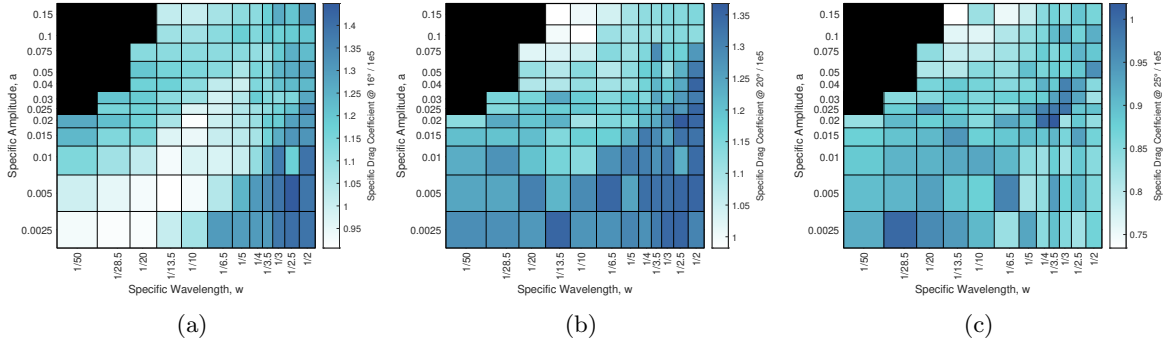


Figure 16: Log-log plots at $Re = 100,000$ of the parameter combinations of specific wavelength, w , and specific amplitude, a showing the specific aerodynamic property C_D/C_{D_0} for a) pre-stall regime of $AoA = 16^\circ$ for $C_{D_0} = 0.16$, b) stall regime of $AoA = 20^\circ$ for $C_{D_0} = 0.23$, and c) post-stall regime of $AoA = 25^\circ$ for $C_{D_0} = 0.41$.

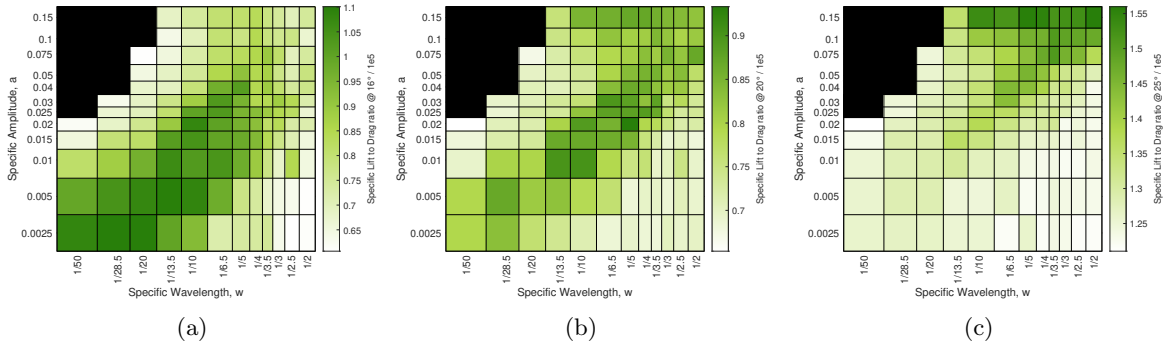


Figure 17: Log-log plots at $Re = 100,000$ of the parameter combinations of specific wavelength, w , and specific amplitude, a showing the specific aerodynamic property $(C_L/C_D)/(C_L/C_D)_0$ for a) pre-stall regime of $AoA = 16^\circ$ for $(C_L/C_D)_0 = 5.20$, b) stall regime of $AoA = 20^\circ$ for $(C_L/C_D)_0 = 3.52$, and c) post-stall regime of $AoA = 25^\circ$ for $(C_L/C_D)_0 = 1.51$.

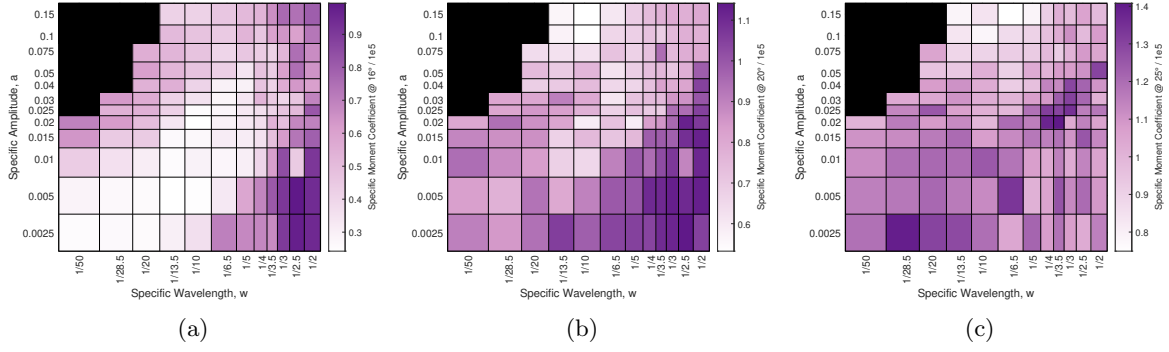


Figure 18: Log-log plots at $Re = 100,000$ of the parameter combinations of specific wavelength, w , and specific amplitude, a showing the specific aerodynamic property C_M/C_{M_0} for a) pre-stall regime of $AoA = 16^\circ$ for $C_{M_0} = 0.12$, b) stall regime of $AoA = 20^\circ$ for $C_{M_0} = 0.12$, and c) post-stall regime of $AoA = 25^\circ$ for $C_{M_0} = 0.11$.

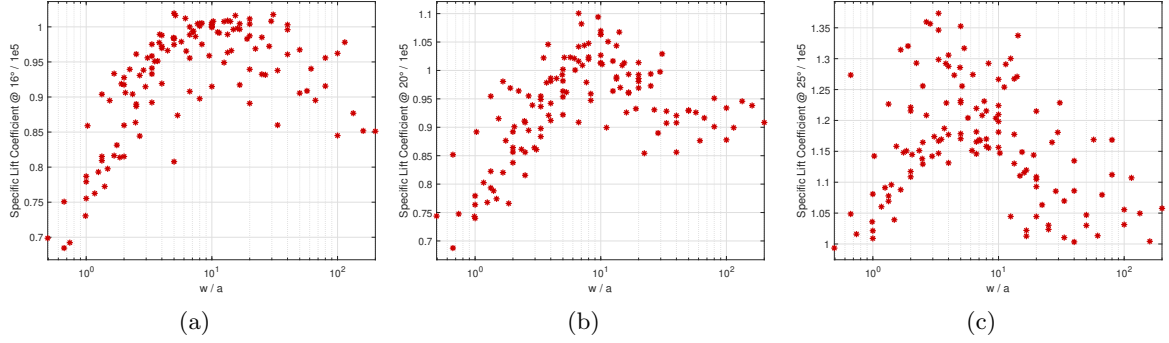


Figure 19: Log plots at $Re = 100,000$ of the aerodynamic property C_L/C_{L_0} against the geometric ratio w/a for a) pre-stall regime of $AoA = 16^\circ$ for $C_{L_0} = 0.83$, b) stall regime of $AoA = 20^\circ$ for $C_{L_0} = 0.81$, and c) post-stall regime of $AoA = 25^\circ$ for $C_{L_0} = 0.62$.

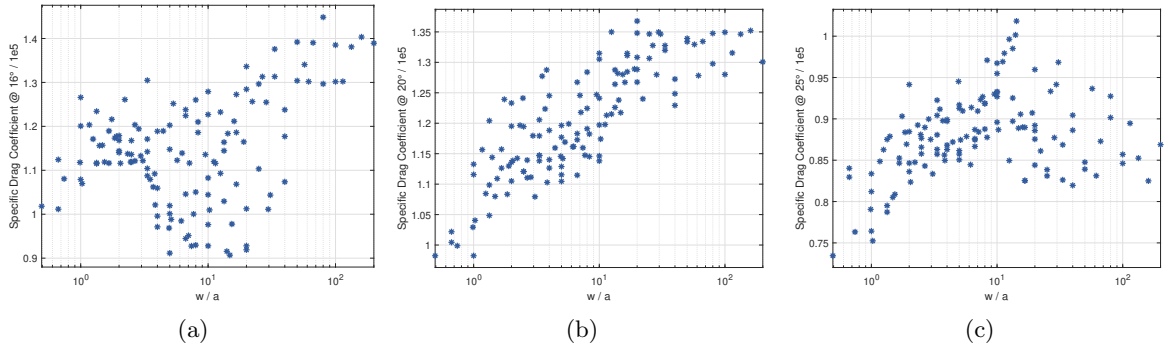


Figure 20: Log plots at $Re = 100,000$ of the aerodynamic property C_D/C_{D_0} against the geometric ratio w/a for a) pre-stall regime of $AoA = 16^\circ$ for $C_{D_0} = 0.16$, b) stall regime of $AoA = 20^\circ$ for $C_{D_0} = 0.23$, and c) post-stall regime of $AoA = 25^\circ$ for $C_{D_0} = 0.41$.

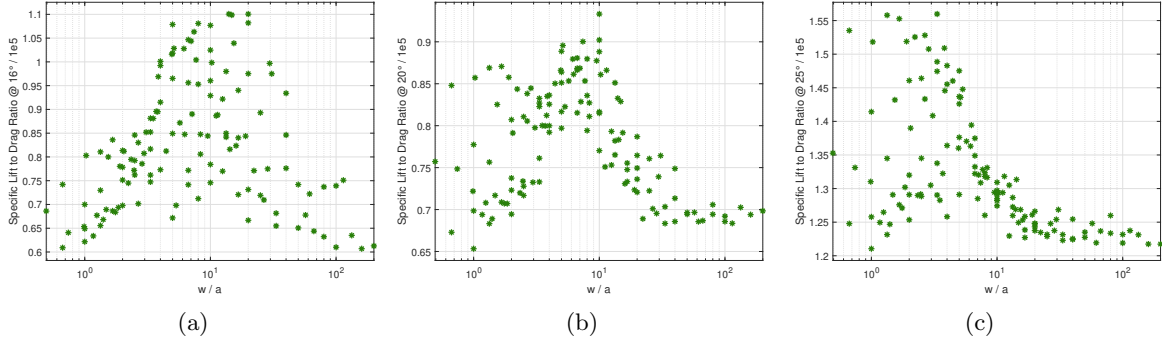


Figure 21: Log plots at $Re = 100,000$ of the aerodynamic property $(C_L/C_D)/(C_L/C_D)_0$ against the geometric ratio w/a for a) pre-stall regime of $AoA = 16^\circ$ for $(C_L/C_D)_0 = 5.20$, b) stall regime of $AoA = 20^\circ$ for $(C_L/C_D)_0 = 3.52$, and c) post-stall regime of $AoA = 25^\circ$ for $(C_L/C_D)_0 = 1.51$.

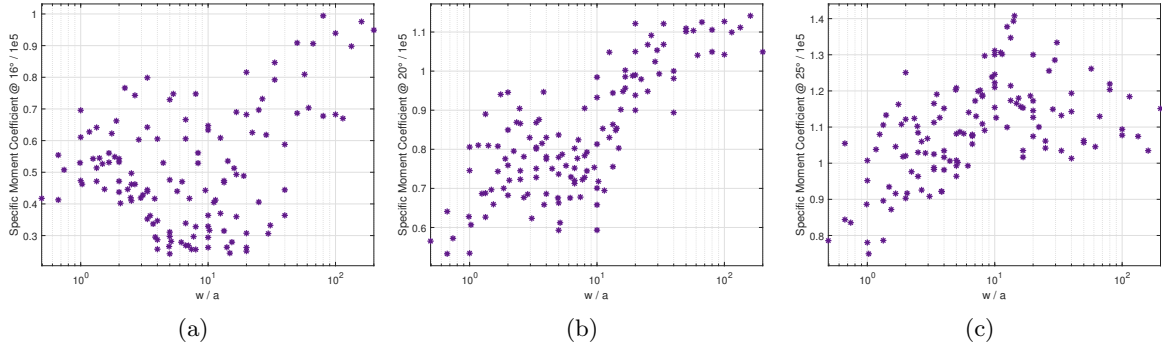


Figure 22: Log plots at $Re = 100,000$ of the aerodynamic property C_M/C_{M_0} against the geometric ratio w/a for a) pre-stall regime of $AoA = 16^\circ$ for $C_{M_0} = 0.12$, b) stall regime of $AoA = 20^\circ$ for $C_{M_0} = 0.12$, and c) post-stall regime of $AoA = 25^\circ$ for $C_{M_0} = 0.11$.

serrations dominates the C_L/C_D values, significantly reducing the performance of the wing sections. This suggests that pre-stall performance is determined by the serrations' ability to reduce drag and to minimally perturb the wing's flow, not by their lift-enhancing ability. As seen in Figure 17a, the peak values are again lying between 8 – 20 with a much thinner peak than that seen in the lift w/a distribution. The majority of the aerodynamic values for serration combinations do not exceed those of the unserrated wing-section pre-stall. This follows the trends seen in previous literature, which state that LE-serrations are stall- and post-stall performance-enhancing devices. The least perturbing serrations at the bottom left-hand corner of the Figure 15a interestingly perform the worst of the entire parameter space, indicating that perturbing the flow without significant serration size seems to break the attached flow seen in the unserrated wing-section without introducing the benefits seen in the other regimes.

The parametric optimisation at stall of 20° is found in Figure 15b, 16b, and 17b. As seen in the experimental validation and following trends of previous literature, LE-serrations mitigate sudden stall-onset. The 20° values are indicative of the performance of the stall-transient region observed in Figure 14, where the unserrated wing-section's sudden stall onset would occur. The data for 20° are less clear for both lift and drag than those for pre-stall. The lift values seen in Figure 15b

follow similar trends to those observed pre-stall but with a significant reduction in "island" width, indicating the performance of LE-serrations around stall is property dependent. The performance trend in lift is still following the pre-stall trends with a performance increase for an estimated gradient w/a between 5 and 7.5 starting at $[1/50, 0.0025, 0.35]_{SC}$. The difference is that larger amplitudes are favoured around stall, indicating that stall-flow is sensitive to serration amplitude. Lift performance is maintained through all wavelengths with the trend terminating at $[1/2, 0.075, 0.35]_{SC}$. The stall drag values seen in 16b favour the high-amplitude serrations significantly, with a significant increase in drag unanimously for all $a < 0.01$ and most low-amplitude, low-wavelength serration combinations seen at the bottom right of the plot. This results in a reduction in C_L/C_D seen in Figure 17b of low-amplitude serrations dominated by the increase in drag. The C_L/C_D has a distinct "island" of increased performance surrounding $[1/6.5, 0.02, 0.35]_{SC}$ which lightly extends down to the extrema following the line of constant $w/a \simeq 5.5$. This suggests a strong correlation between serration properties and stall performance, tightening the high-performing combinations relative to pre-stall and shifting trends towards higher-amplitude, more flow-perturbing serrations. No serration combination's performance exceeds that of the unserrated wing section at stall. The smallest reduction in lift on drag ratio is 5.2% for $[1/6.5, 0.03, 0.35]_{SC}$. This follows trends reported in previous literature, which indicate that LE-serrations mitigate the sudden onset of stall but reduce $C_{L_{max}}$ minimally [12].

The post-stall regime of 25° seen in Figure 15c, 16c, and 17c significantly tightens the "island" of high-performance serration combinations, with favouring of the largest serration amplitudes that are simulated in this study. Whereas the trends in lift and drag performance seen in Figure 15c and 16c, respectively, are much noisier than pre-stall and stall parameter optimisation data, the relative performance of the serrations post-stall seen in the C_L/C_D values in Figure 17c is clear. Almost all low-fidelity serration combinations hover around a 30% increase in lift-to-drag ratio post-stall, with the majority of high-performing wing sections having medium-low wavelengths and high amplitudes. This indicates a strong correlation between serration properties and performance, suggesting that the majority of the effect of creating stream-wise vortices occurs post-stall. Despite the majority of high-performing serration combinations falling at high amplitudes, all serrated combinations studied exceed the performance of the unserrated airfoil with a maximum performance increase in lift to drag ratio of 55% for $[1/5, 0.15, 0.35]_{SC}$. The most interesting result seen in the post-stall regime is the "tail" of high-performing serration combinations that falls down the previously indicated line of $w/a \simeq 5.5$ from the "island" seen in the top right of the plot, which ends at $[1/6.5, 0.03, 0.35]_{SC}$.

The main result from the parametric optimisations is that the performance of LE-serrated wings is highly dependent on the regime in which the wing section is placed. For most applications, the slight reduction in performance pre-stall and at stall is enough to deny the use of LE-serrations on wings. However, for aircraft that nominally operate at high AoAs, their use could be justified. To mitigate pre-stall and stall performance reductions, the most obvious result of this optimisation is to choose the serration combination that minimises pre-stall performance reduction while maintaining post-stall vortex generation and flow separation delay. The serration combination observed at the termination of the "tail" in Figure 17c, $[1/6.5, 0.03, 0.35]_{SC}$, is the most promising specific serration combination as it is contained in all three high-performing "island" regions. For aircraft that wish to explore the use of LE-serrations, the recommendation is to start at the indicated optimal configuration and work

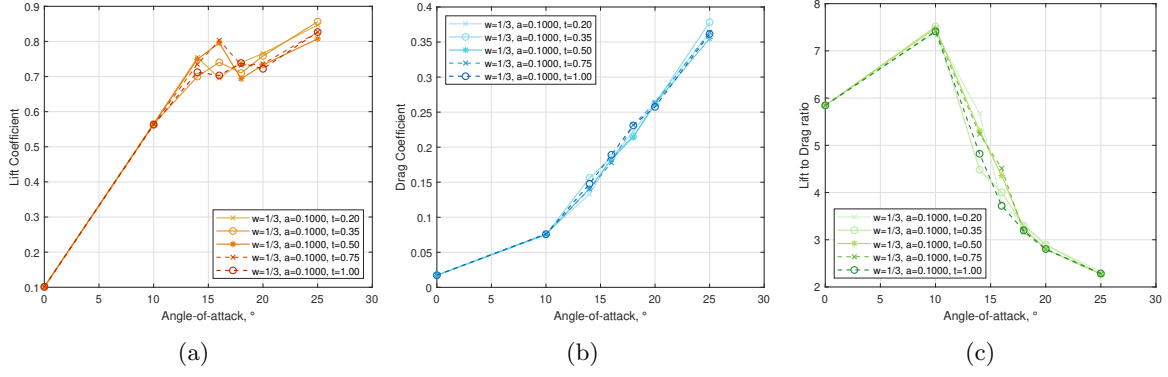


Figure 23: Plots of the numerical data of $[1/3, 0.1, -]_{SC}$ at $Re = 100,000$ for varying termination distances of 0.2, 0.35, 0.5, 0.75, and 1.0 for a range of AoAs, for the aerodynamic properties a) C_L , b) C_D , and c) C_L/C_D .

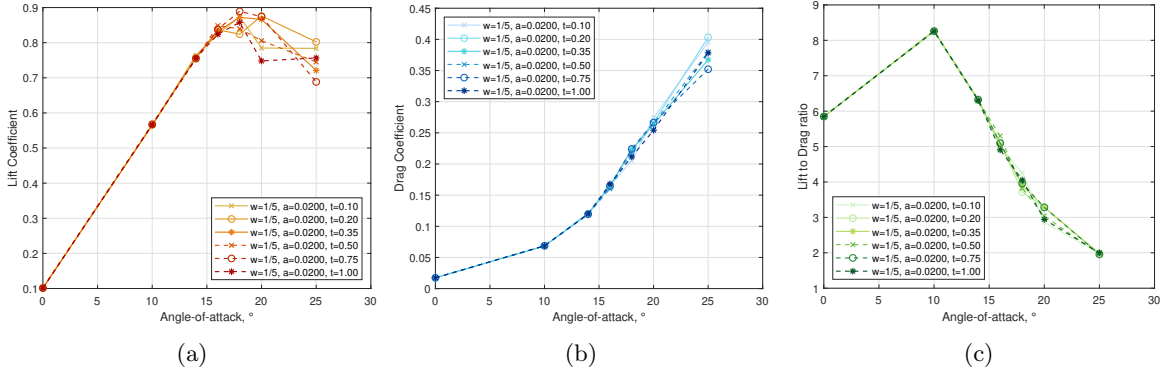


Figure 24: Plots of the numerical data of $[1/5, 0.02, -]_{SC}$ at $Re = 100,000$ for varying termination distances of 0.1, 0.2, 0.35, 0.5, 0.75, and 1.0 for a range of AoAs, for the aerodynamic properties a) C_L , b) C_D , and c) C_L/C_D .

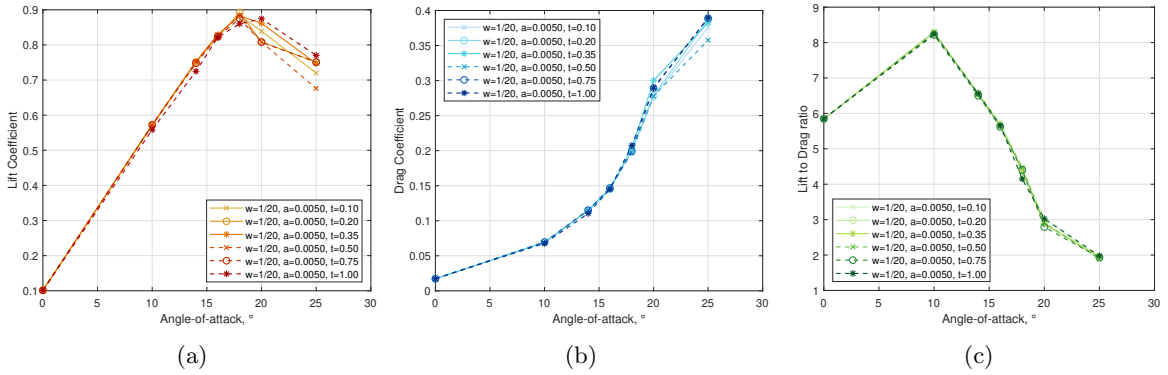


Figure 25: Plots of the numerical data of $[1/20, 0.005, -]_{SC}$ at $Re = 100,000$ for varying termination distances of 0.1, 0.2, 0.35, 0.5, 0.75, and 1.0 for a range of AoAs, for the aerodynamic properties a) C_L , b) C_D , and c) C_L/C_D .

up or down the line of constant $w/a \simeq 5.5$, depending on the specific use case. The results from this section are promising for the future usage of LE-serrations and are a fantastic starting point for a specific case investigation in industry.

4.2.2 Termination Distance

The final geometric property that is parametrised is the termination distance of the serration dimensioned in Figure 3. The main perturbing surface of the LE-serrations is at the LE of the wing, but the termination distance modulates how far the serration channels go back along the chord of the wing. This could affect performance and, most importantly, affect the cost of manufacturing the wings. For small values of t , the section of the wing affected by the serration is minimal, thereby minimising manufacturing costs.

The data for high-amplitude, high-wavelength serration combination $[1/3, 0.1, -]_{SC}$ are seen in Figure 23a, 23b, and 23c for varying values of termination distance. The most obvious result is that the termination distance is much less sensitive to aerodynamic performance. The main region of effect of the termination distance on the performance of the wing section is around stall. The low termination distances $t = 0.2$ and 0.35 seem to favour the post-stall regime, with values maximising at 25° and a reduction in stall and pre-stall values. This result is also observed in Figure 24a, 24b, and 24c, which explores a medium fidelity serration combination of $[1/5, 0.02, -]_{SC}$. The lower termination distance improves post-stall performance with minimal reduction in pre-stall performance, except at the lowest value of 0.1 , which shows significant pre-stall performance reductions. This is intuitive: the shorter the termination distance, the less smooth the transition between the serrated and unperturbed sections of the wing. Whereas lower termination distances are cheaper to manufacture and favour post-stall applications, terminations that are too small result in significant performance reductions.

The least difference in performance across termination distance is that of the high-fidelity serration combination $[1/20, 0.005, -]_{SC}$ seen in Figure 25a, 25b, and 25c. There is a small performance delta in the post-stall regime for lift, suggesting that the smaller the serration perturbations, the less the serration insertion affects performance. Whereas the performance of LE-serrated wings is much less sensitive to the termination distance of the serrations, minor optimisations must be considered when deciding how little the serration perturbs the wing through its termination distance, t .

4.2.3 Vortex Qualification

Although quantitative aerodynamic data are indicative of LE-serration effectiveness, understanding the phenomena underlying the variable effect of serration properties is important. In Figure 26, the vorticity flow characteristics vary wildly with serration geometry. In Figure 26a for the large-serration combination $[1/5, 0.075, 0.35]_{SC}$, the stable vortices are held distinctly to 30% of the chord with minor vortex channels observed 45% of the chord. Looking at the smaller serrations of $[1/10, 0.05, 0.35]_{SC}$ in Figure 26b, the vortex structures are evident at 15% chord, they are minimally present at 30% chord and nonexistent past that. This is worsened when looking at the small serrations of $[1/20, 0.02, 0.35]_{SC}$ in Figure 26c, where there is no sign of any vortex structures past 15% chord, resembling the flow structure of the unserrated wing-section seen in Figure 26d. The serration combination in Figure 26a is part of the high-performing “island” of the post-stall regime parameter optimisation seen in Figure

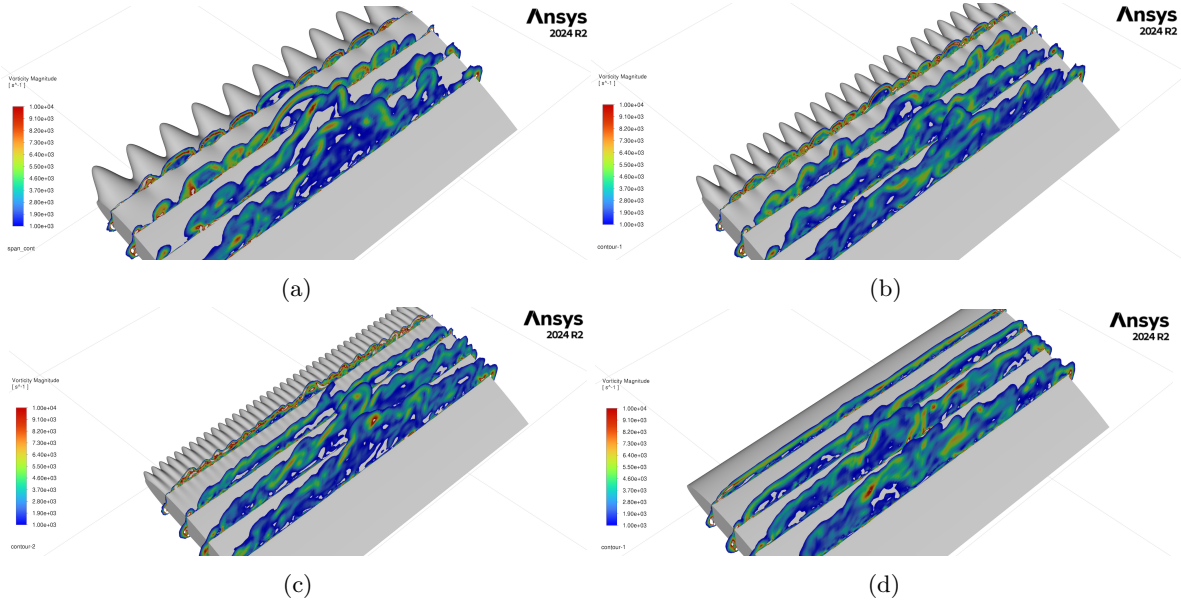


Figure 26: Vorticity magnitude plots at $Re = 100,000$ of planar surfaces 15 mm, 30 mm, 45 mm, and 60 mm from the LE at 25° for serration combinations of a) $[1/5, 0.075, 0.35]_{SC}$, b) $[1/10, 0.05, 0.35]_{SC}$, c) $[1/20, 0.02, 0.35]_{SC}$, and d) unserrated.

17c, and the persistence of the vortex structures past 50% of the chord is a promising reason why. The total lack of stable structures in the higher-fidelity serrations is correlated with their reduction in performance post-stall. It presents a strong argument for the causation of LE-serration performance and the generation and maintenance of stream-wise vortices in a wing-section’s aerodynamic performance.

4.3 High Reynolds Number

The high Re investigation will be conducted at $Re = 5 \times 10^5$ with a fluid velocity of $V_H = 73.04 \text{ m s}^{-1}$ as described in Section 3.2.1. As the EWT cannot reach the required velocity, the determination of the pre-stall, stall, and post-stall regimes will be performed purely computationally. In Figure 27 are the stall distributions of 3 serration combinations including the unserrated airfoil. The value for pre-stall will be taken as 23° , stall as 29° and post-stall 34° . While the post-stall regime of 34° is rarely observed in aircraft, this study seeks to identify the stall characteristics of serrated airfoils that are invariant across aircraft.

4.3.1 Wavelength and Amplitude

The value of termination distance, t , again will be held constant at the value 0.35 for the initial optimisation, and the variable effect of termination distance will be explored in Section 4.3.2.

The pre-stall trends at 23° observed in Figures 28a, 29a, and 30a are extremely clear with again a wide “island” of well performing serration combinations that extends from the lower right, to the upper left terminating a value of $w/a = 9$ best seen in Figure 32a. Unlike the low-Reynolds number data, the serration combinations at the highest value of a/w , the closest values to the unserrated airfoil, are

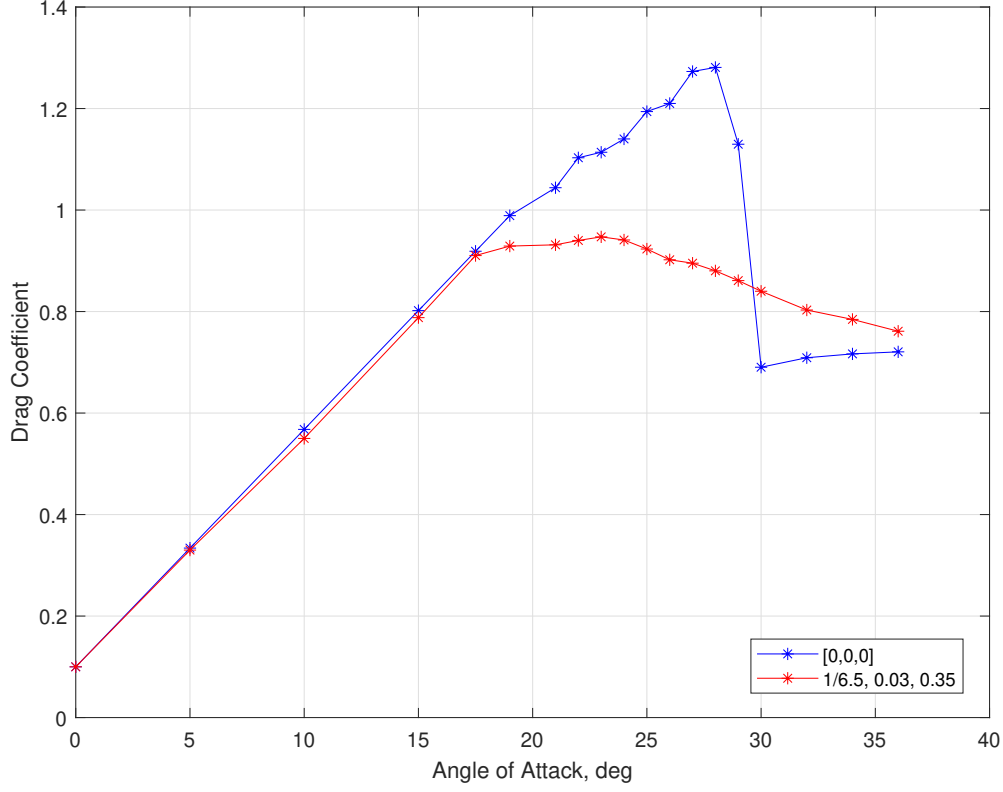


Figure 27: Plot of the stall profiles at $Re = 500,000$ of the unserrared airfoil, the $[1/28.5, 0.015, 0.35]_{SC}$ and the $[1/28.5, 0.015, 0.35]_{SC}$ giving the pre-stall regime of 23° , stall regime of 29° and post-stall regime of 34° .

contained within the high-performing “island”. This difference suggests that at higher velocities, the slight perturbations caused by the low-fidelity serrations are “pushed” past and don’t destroy the pre-stall characteristics. At $Re = 100,000$, this low performance seen in Figures 15a and 19a is indicative of the sensitivity of the stall profile which is lost when moving to higher Re . There is an obvious “notch” seen cut out of the high-performing serration combinations which starts at $[1/5, 0.02, 0.35]_{SC}$ and moves up at a line of constant $w/a = 22$. The drag values seen in Figures 29a and 33a are again bi-model with majority of the high-drag serration combinations occurring at high-wavelength and high-amplitudes. The lowest drag values occur at the highest and the lowest values of w/a with exceptions near $[1/28.5, 0.015, 0.35]_{SC}$ and $[1/4, 0.04, 0.35]_{SC}$ which is similarly observed at the lower Re . The cut-off between the high and low performing serration combinations is seen to be much more stark when looking at the plots for lift to drag ratio in Figures 30a and 34a with a steep drop off in performance past the w/a value of 10. At higher Re , it seems the regions of high-performing serration combinations are much more defined than that observed at the lower-Reynolds number.

The stall trends at 29° observed in Figures 28b, 29b, and 30b significantly favour the low-fidelity serrations as the majority of the high lift values as observed in Figure 32b occur at w/a values

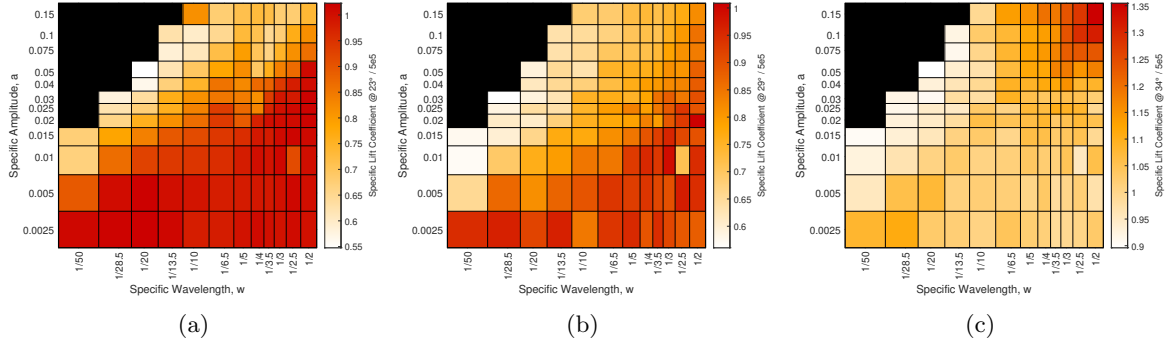


Figure 28: Log-log plots at $Re = 500,000$ of the parameter combinations of specific wavelength, w , and specific amplitude, a showing the specific aerodynamic property C_L/C_{L_0} for a) pre-stall regime of $AoA = 23^\circ$ for $C_{L_0} = 1.11$, b) stall regime of $AoA = 29^\circ$ for $C_{L_0} = 1.13$, and c) post-stall regime of $AoA = 34^\circ$ for $C_{L_0} = 0.72$.

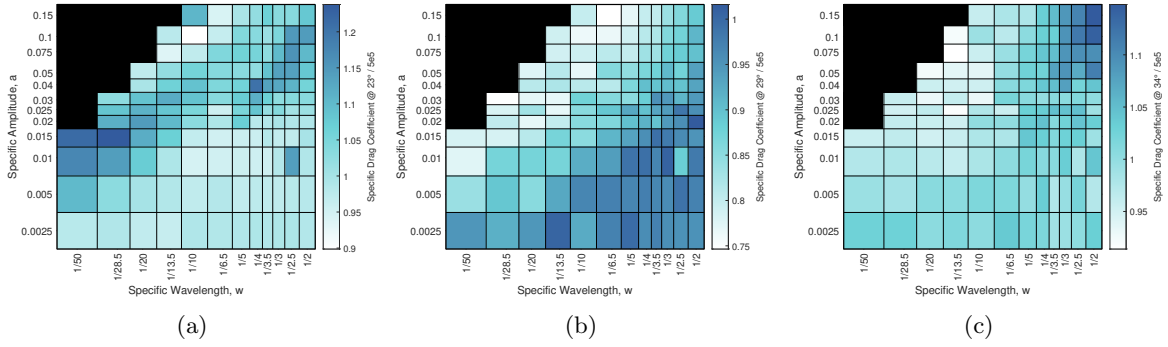


Figure 29: Log-log plots at $Re = 500,000$ of the parameter combinations of specific wavelength, w , and specific amplitude, a showing the specific aerodynamic property C_D/C_{D_0} for a) pre-stall regime of $AoA = 23^\circ$ for $C_{D_0} = 0.29$, b) stall regime of $AoA = 29^\circ$ for $C_{D_0} = 0.51$, and c) post-stall regime of $AoA = 34^\circ$ for $C_{D_0} = 0.49$.

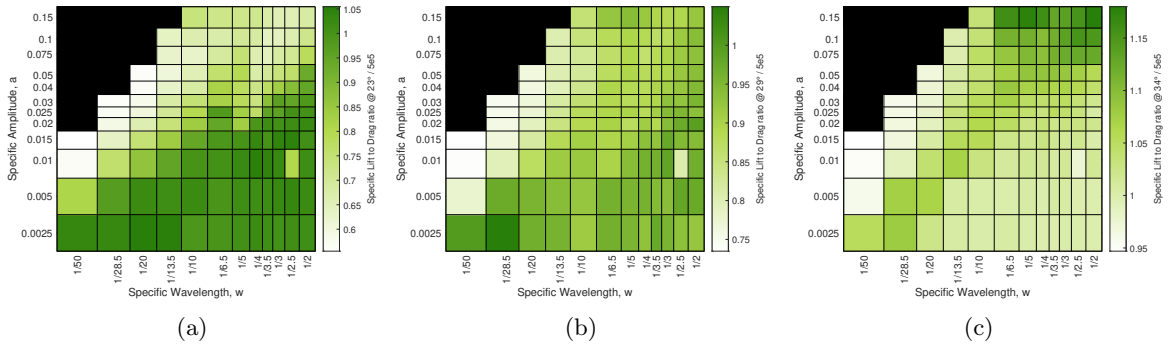


Figure 30: Log-log plots at $Re = 500,000$ of the parameter combinations of specific wavelength, w , and specific amplitude, a showing the specific aerodynamic property $(C_L/C_D)/(C_L/C_D)_0$ for a) pre-stall regime of $AoA = 23^\circ$ for $(C_L/C_D)_0 = 3.82$, b) stall regime of $AoA = 29^\circ$ for $(C_L/C_D)_0 = 2.21$, and c) post-stall regime of $AoA = 34^\circ$ for $(C_L/C_D)_0 = 1.46$.

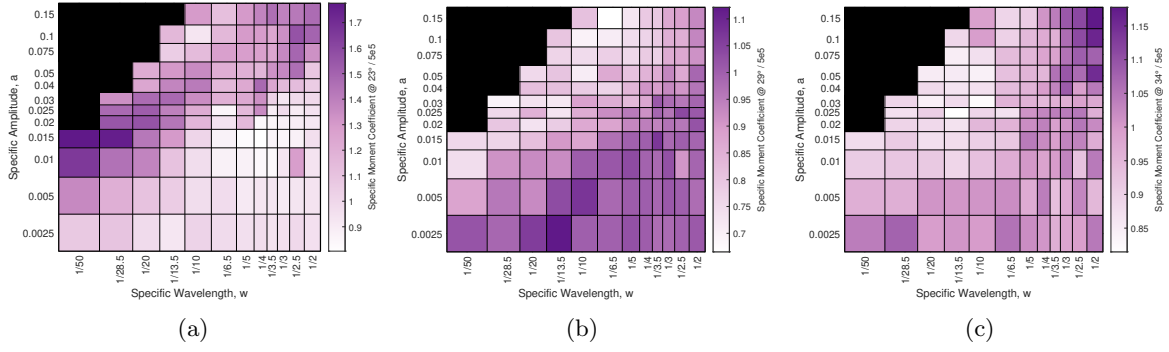


Figure 31: Log-log plots at $Re = 500,000$ of the parameter combinations of specific wavelength, w , and specific amplitude, a showing the specific aerodynamic property C_M/C_{M_0} for a) pre-stall regime of $AoA = 23^\circ$ for $C_{M_0} = 0.077$, b) stall regime of $AoA = 29^\circ$ for $C_{M_0} = 0.15$, and c) post-stall regime of $AoA = 34^\circ$ for $C_{M_0} = 0.13$.

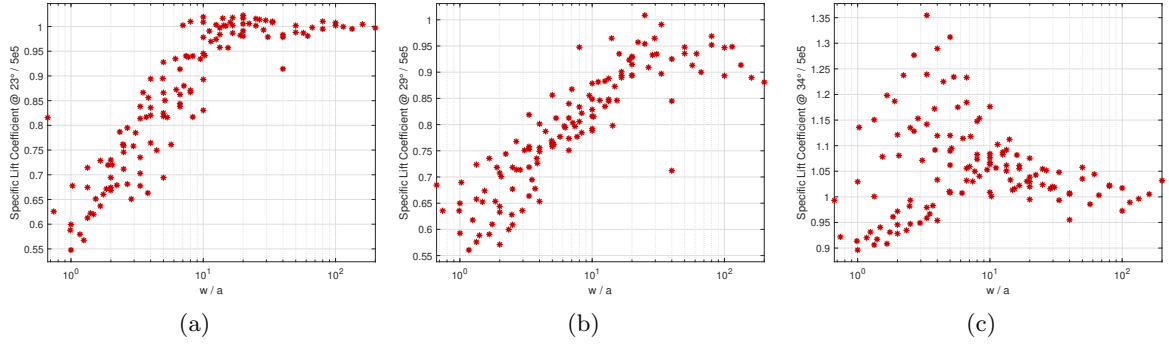


Figure 32: Log plots at $Re = 500,000$ of the aerodynamic property C_L/C_{L_0} against the geometric ratio w/a for a) pre-stall regime of $AoA = 23^\circ$ for $C_{L_0} = 1.11$, b) stall regime of $AoA = 29^\circ$ for $C_{L_0} = 1.13$, and c) post-stall regime of $AoA = 34^\circ$ for $C_{L_0} = 0.72$.

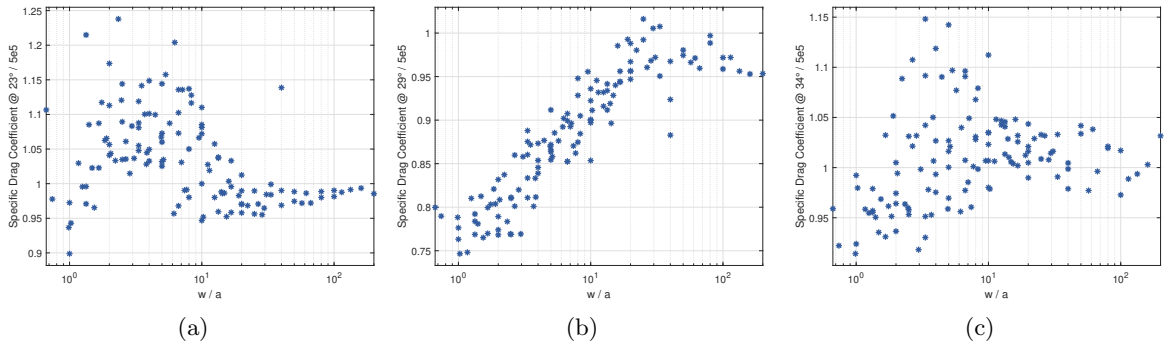


Figure 33: Log plots at $Re = 500,000$ of the aerodynamic property C_D/C_{D_0} against the geometric ratio w/a for a) pre-stall regime of $AoA = 23^\circ$ for $C_{D_0} = 0.29$, b) stall regime of $AoA = 29^\circ$ for $C_{D_0} = 0.51$, and c) post-stall regime of $AoA = 34^\circ$ for $C_{D_0} = 0.49$.

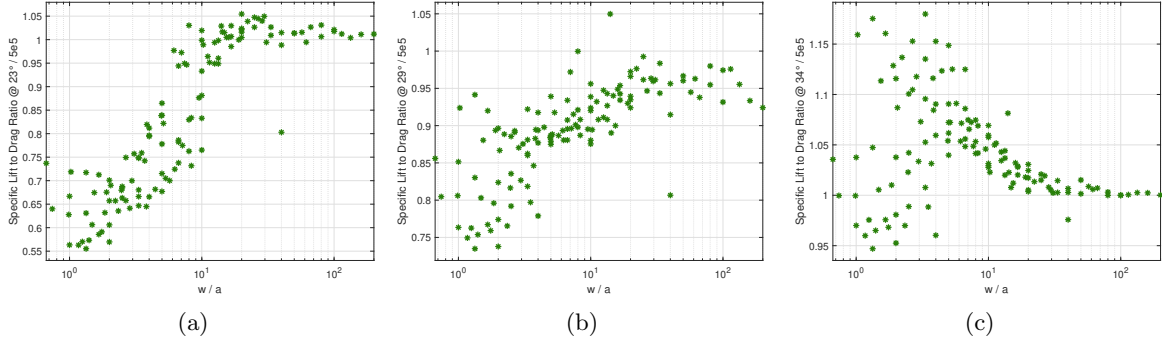


Figure 34: Log plots at $Re = 500,000$ of the aerodynamic property $(C_L/C_D)/(C_L/C_D)_0$ against the geometric ratio w/a for a) pre-stall regime of $AoA = 23^\circ$ for $(C_L/C_D)_0 = 3.82$, b) stall regime of $AoA = 29^\circ$ for $(C_L/C_D)_0 = 2.21$, and c) post-stall regime of $AoA = 34^\circ$ for $(C_L/C_D)_0 = 1.46$.

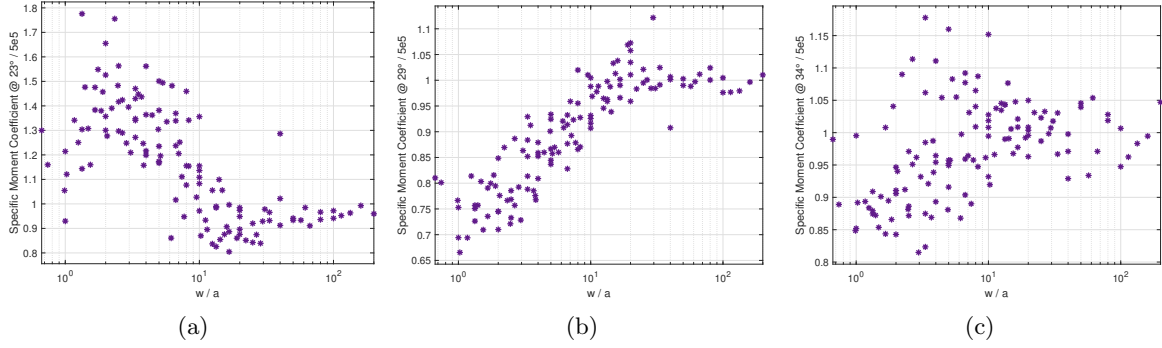


Figure 35: Log plots at $Re = 500,000$ of the aerodynamic property C_M/C_{M_0} against the geometric ratio w/a for a) pre-stall regime of $AoA = 23^\circ$ for $C_{M_0} = 0.077$, b) stall regime of $AoA = 29^\circ$ for $C_{M_0} = 0.15$, and c) post-stall regime of $AoA = 34^\circ$ for $C_{M_0} = 0.13$.

of 15 and above. The notable exceptions are the serration combinations that of the hover around $[1/28.5, 0.025, 0.35]_{SC}$ with w/a values closer to 8–10. Interestingly, around stall for $Re = 5 \times 10^5$, the drag values are the highest at higher values of w/a seen in Figure 33b, the opposite trend to that of lift. The lowest values for drag occur at the highest explored values of w/a , with an almost 25% reduction in drag comparing $[1/2.5, 0.05, 0.35]_{SC}$ and $[1/28.5, 0.03, 0.35]_{SC}$. While most aircraft optimise for general performance, the inverse trends suggests that at higher Re , if an aircraft is operating near stall, the main effect is their ability to reduce drag, not as a lift device. The observed lift to drag ratio in Figure 30b is the flattest distribution observed with the only high-performing serration combinations occurring near $[1/28.5, 0.0025, 0.35]_{SC}$. This is also observed in the w/a distribution plot in Figure 30b with a flat trend passing through the middle of the distribution.

The post-stall trends at 34° observed in Figures 28c, 29c, and 30c are very similar to that observed at the lower Re . The performance of the low-amplitude serration combinations completely diminish and the highest amplitude serration combinations dominate. The peaks of the w/a distributions seen in Figures 32c, 33c, and 34c switch from high values to low when moving through stall. When looking at the lift to drag ratio post-stall in Figure 30c, similar to that of the low Re , a “tail” of high performing serration combinations extends past down into the low amplitude combinations following the line of

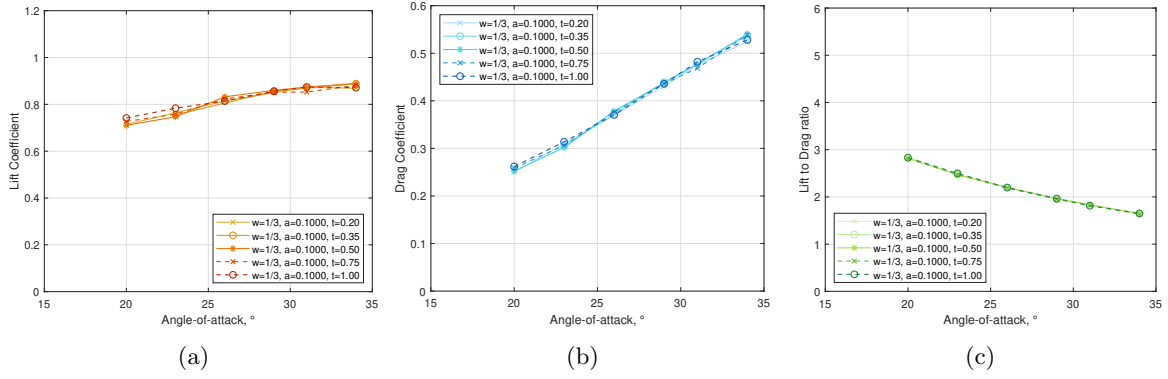


Figure 36: Plots of the numerical data of $[1/3, 0.1, -]_{SC}$ at $Re = 500,000$ for varying termination distances of 0.2, 0.35, 0.5, 0.75, and 1.0 for a range of AoAs, for the aerodynamic properties a) C_L , b) C_D , and c) C_L/C_D .

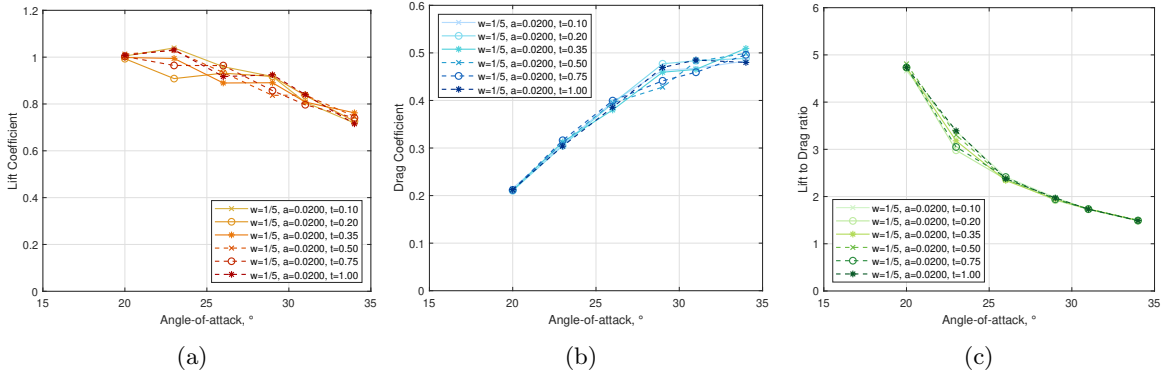


Figure 37: Plots of the numerical data of $[1/5, 0.02, -]_{SC}$ at $Re = 500,000$ for varying termination distances of 0.1, 0.2, 0.35, 0.5, 0.75, and 1.0 for a range of AoAs, for the aerodynamic properties a) C_L , b) C_D , and c) C_L/C_D .

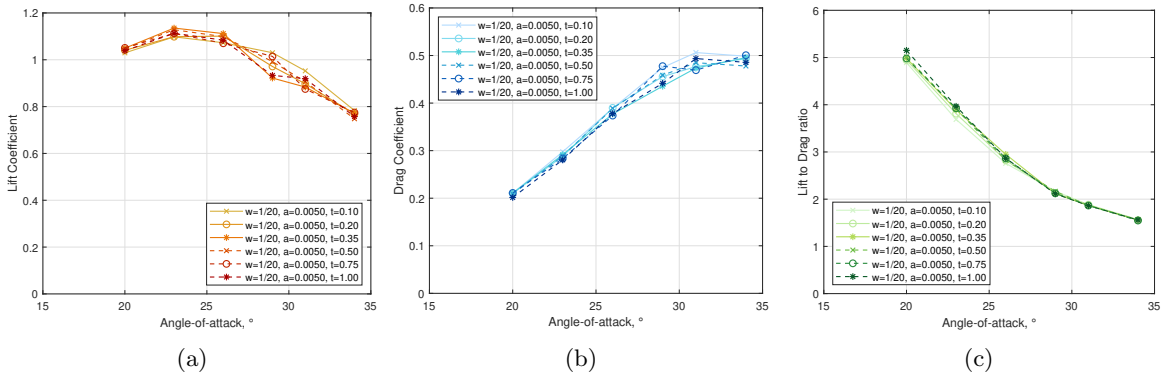


Figure 38: Plots of the numerical data of $[1/20, 0.005, -]_{SC}$ at $Re = 500,000$ for varying termination distances of 0.1, 0.2, 0.35, 0.5, 0.75, and 1.0 for a range of AoAs, for the aerodynamic properties a) C_L , b) C_D , and c) C_L/C_D .

constant $w/a = 5.5$. In the high Re regime, there is no evident optimal configuration but the same trends seen in the low Re regime emerge. It suggests that at higher Re , the ability for serrated leading edges to perform well in all 3 regimes reduces. For a specific industry investigation into LE-serrations above $Re = 500,000$, the recommendation for LE-serration would only be given if the wing is targeting performance increase in a single stall regime.

4.3.2 Termination Distance

The data for high-amplitude, high-wavelength serration combinations $[1/3, 0.1, -]_{SC}$, $[1/5, 0.02, -]_{SC}$ and $[1/20, 0.005, -]_{SC}$ are seen in Figure 36, 37, and 38 for varying values of termination distance. The most obvious result and one that is shared with the low Re results is that the termination distance is much less sensitive to aerodynamic performance. The main region of effect of the termination distance on the performance of the wing section is around stall best observed the serration combinations $[1/5, 0.02, -]_{SC}$ and $[1/20, 0.005, -]_{SC}$ in Figures 37 and 38. The differences between the serration distances pre and post-stall seem to be minimal which is different to that of low Re results. This suggest that the higher velocity flow is less sensitive to the insertion of the serration. This being said, the results agree with the low Re results that the higher terminations distances, the greater the performance. This is observed mostly in $[1/5, 0.02, -]_{SC}$.

The least difference in performance across termination distance is that of the low-fidelity serration combination $[1/20, 0.005, -]_{SC}$ seen in Figure 36. This is the opposite to what is observed in and suggests that at higher Re , the importance of fully inserted serrations increase. Whereas the performance of LE-serrated wings is much less sensitive to the termination distance of the serrations, minor optimisations must be considered when deciding how little the serration perturbs the wing through its termination distance especially if you are designing high-fidelity serrations at higher Re .

4.3.3 Vortex Qualification

In Figure 39, the vorticity flow characteristics vary wildly with serration geometry. In Figure 39a for the large-serration combination $[1/5, 0.075, 0.35]_{SC}$, the stable vortices are held distinctly to 30% of the chord with minor vortex channels observed 45% of the chord. This is very similar to that of the low Re qualification analysis previously. Looking at the slightly finer serrations in Figure 39b, the vortices are evident at 15% of the chord but are only faintly present at 30% of the chord where at the low Re , these structures were stable further downstream. Comparing to the smaller serrations on $[1/10, 0.05, 0.35]_{SC}$ seen in Figure 39b and on $[1/20, 0.02, 0.35]_{SC}$ in Figure 39c, there is 0 sign of any stable vorte structures past 15% of the chord. This suggests that at higher Re , the ability to form stable vortex structures decreases which reduces the number of optimal serration combinations which is what is observed in Section 4.3.1. When comparing to the unserrated airfoil in Figure 39d, the unserrated airfoil has no stable vortex structures and again, is representative if its reduce performance post-stall.

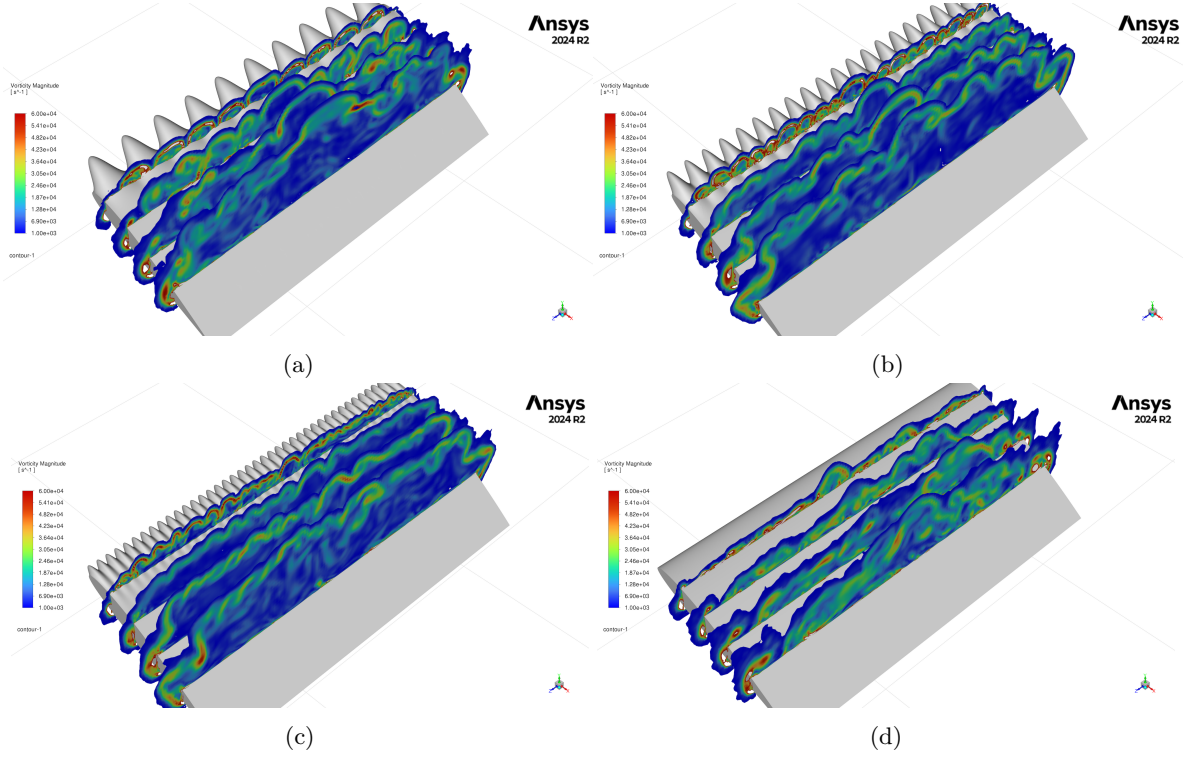


Figure 39: Vorticity magnitude plots at $Re = 500,000$ of planar surfaces 15 mm, 30 mm, 45 mm, and 60 mm from the LE at 34° for serration combinations of a) $[1/5, 0.075, 0.35]_{SC}$, b) $[1/10, 0.05, 0.35]_{SC}$, c) $[1/20, 0.02, 0.35]_{SC}$, and d) unserrated.

5 Conclusion

This report successfully identifies the variable effect of the geometric properties of LE-serrations on the aerodynamics of wing sections. The pre-stall, stall, and post-stall regimes were each explored, and the optimal configurations were identified. The parameter space regions and trends of optimal performance result in the most general optimal configuration at $Re = 100,000$ for all regimes of $[1/6.5, 0.03, 0.35]_{SC}$. A strong correlation between the ratio $w/a \simeq 5.5$ and aerodynamic performance is observed across all stall regimes for both Re explored. The most evident trend is that, in lower-AoA regimes, less flow-perturbing serrations perform best, whereas greater perturbing serrations perform best at high-AoA regimes, with the generation and maintenance of streamwise stable vortex structures. The termination distance of the serrations has a minimal effect on aerodynamics, except for small insertions, which sharpen the transition from the serrations to the body and reduce performance. A value of $t = 0.2 - 0.35$ is recommended as a starting point to then optimise for slight aerodynamic metrics and manufacturing costs. The recommended future work on LE-serration properties includes exploring higher-amplitude, higher-wavelength configurations, higher Re , and different airfoil profiles for different aircraft applications.

For aircraft that wish to explore the use of LE-serrations, the results from this report are auspicious and serve as a fantastic starting point for a specific case investigation in industry.

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